



Porosity formation and suppression mechanisms in oscillating laser-arc hybrid welding of aluminum alloys



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Highlights

- Compared with LAHW, the dynamic heat input induced by beam oscillation in OLAHW, together with the enhanced heat transfer within the molten pool, significantly delays the accumulation of vapor pressure inside the keyhole, thereby extending the time required for stable keyhole formation by approximately 3–4 times.

- In OLAHW, porosity formation is mainly attributed to the vortex-driven closure effect and the impact effect associated with droplet transfer.
- Three typical porosity suppression mechanisms in OLAHW of aluminum alloys were elucidated: vortex-driven accelerated bubble escape, keyhole rotation-induced bubble fusion and absorption, and beam oscillation-enhanced keyhole stability. The overall suppression effect is governed by the competition among these mechanisms under different oscillating parameters.

Abstract

Effective suppression of porosity remains a key technical challenge for achieving high-quality fusion welding of high-strength aluminum alloys. Oscillating laser-arc hybrid welding (OLAHW) has attracted increasing attention due to its remarkable porosity-suppression capability; however, systematic investigations on the molten-pool thermo-fluid dynamics and the underlying mechanisms of pore formation and mitigation are still limited. In this work, a numerical model of OLAHW for AA6061 aluminum alloy is established to elucidate the dynamic behaviors of laser keyhole evolution and pore formation, and to clarify the pore-suppression mechanisms induced by beam oscillation. The results show that increasing the oscillating radius r from 0 to 1.0 mm reduces the porosity fraction from 6.64 % to 0.61 % (a 90.8 % reduction), and raising the oscillating frequency f from 100 Hz to 500 Hz completely eliminates porosity. Pore formation is primarily attributed to keyhole instability and collapse induced by vortex flow and droplet impingement. Beam oscillation suppresses porosity through three mechanisms: vortex-driven acceleration of bubble escape, keyhole rotational motion enabling bubble capture and coalescence, and enhanced keyhole stability that prevents collapse. The overall suppression effectiveness is governed by the competition among these mechanisms under different oscillating parameters. This study provides a comprehensive understanding of the dynamic mechanisms of pore formation and suppression in OLAHW, offering theoretical guidance for process optimization and defect control.



Keywords

Numerical simulation; Laser beam oscillation; Laser-arc hybrid welding; Aluminum alloy; Molten pool dynamics; Porosity

1. Introduction

Aluminum alloys, owing to their low density, high specific strength, and excellent corrosion resistance, have become preferred materials for lightweight structural applications and are widely used in modern industries such as aerospace and rail transportation [1,2]. However, their inherent physicochemical properties—such as a coefficient of thermal expansion approximately twice that of steel, a thermal conductivity nearly five times higher than that of steel, and an elastic modulus only one-third that of steel [3,4]—result in poor weldability. In laser welding of aluminum alloys, metallurgical defects, particularly process-induced porosity, remain one of the most critical technical challenges that urgently need to be addressed [5]. To elucidate the formation mechanisms of porosity, extensive experimental and numerical studies have been conducted [[6], [7], [8]]. It is generally accepted that during welding, the laser-induced keyhole is highly unstable, and its root is prone to collapse, generating gas bubbles that may subsequently be captured by the advancing solid–liquid interface during solidification, ultimately forming pores. Owing to this formation mechanism, it is difficult to achieve effective porosity suppression through process optimization alone in conventional laser welding.

Approximately a decade ago, researchers attempted to employ laser–arc hybrid welding (LAHW) to mitigate the high sensitivity of aluminum alloys to porosity defects during laser welding by enhancing molten pool stability, enlarging the molten pool size, and prolonging the molten pool lifetime. Katayama et al. reported that laser–arc hybrid welding can promote bubble escape through the combined effects of arc heat input and arc force, thereby reducing the tendency for porosity formation to a certain extent [9]. To further clarify the formation and evolution mechanisms of porosity in LAHW, Huang et al. developed a three-dimensional multiphase numerical model and demonstrated that clockwise vortices driven by recoil pressure in the lower region of the molten pool can induce keyhole collapse and bubble generation; bubbles that fail to escape in time are eventually captured

by the solidification front and evolve into pores [10]. Xu et al. established a simplified three-dimensional numerical model for aluminum alloy LAHW and showed that appropriately increasing the arc current can enhance keyhole stability and promote bubble flotation, thereby reducing porosity content in the weld [11]. Nevertheless, with further investigations, it has been demonstrated that this approach does not fundamentally resolve the porosity issue [12], [13], [14], and satisfactory suppression effects can only be achieved within a very narrow range of high arc currents under ideal experimental conditions. This limitation primarily arises because pores usually form near the bottom of the molten pool, whereas the penetration depth of the arc heat source is limited and insufficient to remelt the solidifying region and release trapped pores.

To address this challenge, various auxiliary external fields—such as magnetic fields, ultrasonic vibration, mechanical vibration, and optical field modulation—have been introduced over the past decade to suppress porosity during welding [15], [16], [17]. Among them, laser beam oscillation, as a typical implementation of optical field modulation, has attracted increasing attention due to its non-contact nature, rapid response, high controllability, and ease of system integration. It enables direct and active regulation of heat and mass transfer within the molten pool and is therefore considered to possess significant potential for porosity control in aluminum alloy fusion welding [18,19]. In this context, extensive experimental studies have been conducted to explore the effectiveness of oscillating laser welding and oscillating laser-arc hybrid welding in suppressing porosity. For instance, Fetzer et al. employed high-speed in situ X-ray imaging and observed that at a beam oscillating frequency of 100 Hz, bubbles could contact the moving keyhole and escape from the molten pool, whereas at 200 Hz, improved keyhole stability suppressed keyhole collapse and subsequent bubble formation [20]. Liu et al. found that figure-eight oscillating patterns provided stronger molten pool stirring and a larger oscillating range, resulting in effective porosity suppression, although the underlying mechanisms remained unclear [21]. In addition, Wu et al. reported that in oscillating laser-arc hybrid welding (OLAHW) of 7204 aluminum alloy, beam oscillation not only significantly reduced porosity defects but also promoted microstructural refinement, leading to an approximately 81 % improvement in joint ductility compared with non-oscillating conditions [22]. Similar conclusions were also reported by Wang et al. in OLAHW of 2219-T6 aluminum alloy [23]. These studies collectively demonstrate that laser beam oscillation plays a significant role in suppressing porosity defects and improving weld microstructure and mechanical properties.

To gain deeper insight into the molten pool thermofluid behavior and the mechanisms of porosity formation and suppression under beam oscillation, numerous numerical studies on oscillating laser welding have been conducted in recent years. Mohan et al. found that, compared with non-oscillating welding, beam oscillation induces enhanced molten pool stirring and strengthens forced convection, which is a key factor in achieving homogenization of the temperature field [24]. Dakka et al. reported that at

relatively low oscillating frequencies, increasing the oscillating radius can improve the dynamic balance between recoil pressure and surface tension along the keyhole inner wall, thereby suppressing keyhole collapse [25]. Ke et al. also demonstrated that beam oscillation can reduce the temperature and pressure at the rear wall of the keyhole, suppress liquid metal protrusion in the middle and rear regions of the keyhole, decrease keyhole closure tendencies, and hinder bubble generation [26]. Beyond enhancing keyhole stability to suppress bubble formation, Liu et al. further revealed that periodic keyhole motion can actively capture and remove bubbles, thereby reducing porosity formation in the weld [27]. Overall, the mechanisms governing porosity formation and suppression in oscillating laser welding have been systematically investigated, and key effects such as enhanced molten pool fluidity, improved keyhole stability, and bubble capture by oscillating keyholes have been extensively elucidated.

In contrast, in oscillating laser-arc hybrid welding, the simultaneous presence of laser-arc heat source coupling, droplet transfer and impact, and strong vortex flow within the molten pool—beyond laser keyhole dynamics alone—results in a significantly more complex thermofluid coupling behavior. Constrained by such complexity, existing studies on OLAHW have predominantly focused on experimental observations, while systematic numerical investigations of the OLAHW process remain relatively scarce. In recent years, only a limited number of studies have attempted numerical modeling and mechanistic analysis of OLAHW and achieved some representative results. For example, Shi et al. reported that beam oscillation can induce high-speed vortices in the molten pool; when the oscillating frequency increases to 150 Hz, the vortex velocity reaches approximately 0.7 m/s and begins to dominate molten pool flow, stabilizing the disturbances caused by droplet impingement [28]. Wang et al. developed a numerical validation model for spatter behavior in OLAHW based on a self-developed thermal-flow marker method and revealed three typical spatter formation mechanisms under oscillating conditions, as well as their competitive dependence on oscillating parameters [29]. Liao et al. established a high-fidelity numerical model for aluminum alloy OLAHW based on a self-developed VOF discretization scheme and found that oscillation-induced high-speed cyclic flow alters the flow direction of liquid metal in the coupling zone, causing droplet-induced impacts to deviate from the keyhole region at an earlier stage, thereby reducing keyhole instability. Moreover, liquid metal flowing backward along the welding direction transports heat carried by droplets to the rear of the molten pool, further enhancing overall flow stability and uniformity within the molten pool [30].

However, despite the significant advances achieved in the aforementioned OLAHW numerical simulations concerning molten pool flow regulation, spatter behavior, and keyhole stability, a unified and systematic numerical characterization of the formation, evolution, and suppression processes of porosity—the most engineering-sensitive and mechanistically complex metallurgical defect in aluminum alloy OLAHW—remains lacking. Under oscillating laser-arc coupled conditions, the origin mechanisms of porosity, its dynamic evolution pathways, and its response to different oscillating parameters are still not well

understood, and the associated mechanisms require further in-depth investigation. In view of this, the present study establishes a high-fidelity thermofluid coupled numerical model for aluminum alloy OLAHW, specifically aimed at elucidating the mechanisms of porosity formation and suppression. With porosity as the central focus, this work systematically reveals the generation, migration, and suppression mechanisms of pores under oscillating laser scanning conditions and further clarifies the competitive relationships among different porosity suppression mechanisms, thereby providing new theoretical insights for the active control of porosity defects in oscillating laser–arc hybrid welding.

2. Mathematical models

Oscillating laser–arc hybrid welding involves strong coupling among multiple physical fields, including the optical and thermal fields, which leads to highly complex droplet transfer behavior, energy distribution within the molten pool, and melt flow characteristics, thereby posing significant challenges for numerical simulation [31]. Therefore, in order to achieve a balance between computational accuracy and efficiency, the following reasonable assumptions and simplifications are adopted prior to the development of the Flow-3D numerical model(version 11.1):

- (1) The aluminum alloy substrate is assumed to be isotropic, and the absorption coefficients for both laser and arc energy are taken as constants;
- (2) Arc droplets are introduced into the computational domain in the form of spherical momentum sources to represent the droplet transfer process;
- (3) The effects of plasma and arc plume on the welding process are neglected;
- (4) The molten metal within the molten pool is treated as an incompressible Newtonian laminar fluid.

2.1. Heat source model

In this study, a three-dimensional rotating Gaussian volumetric heat source model and a double-ellipsoidal arc heat source model are employed to describe the energy input of the laser and the arc into the molten pool, respectively. The corresponding

heat source formulations are given in Eq. (1) [32] and Eq. (2) [33].

$$q_1(x, y, z) = \frac{3c_s \eta_1 P}{\pi H(1-e^{-3})} \exp \left[\frac{-3c_s(x^2+y^2)}{\log\left(\frac{H}{z}\right)} \right] \quad (1)$$

$$\begin{cases} q_f(x, y, z) = \frac{6\sqrt{3}f_f\eta_2 UI}{a_1 b c \pi \sqrt{\pi}} \exp \left(-\frac{3x^2}{a_1^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2} \right), & x > 0 \\ q_r(x, y, z) = \frac{6\sqrt{3}f_r\eta_2 UI}{a_2 b c \pi \sqrt{\pi}} \exp \left(-\frac{3x^2}{a_2^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2} \right), & x < 0 \end{cases} \quad (2)$$

Where P is the laser power, η_1 is the total absorptivity of the material for the laser heat source, H is the heat source height, and c_s is the heat concentration coefficient. U and I represent the arc voltage and welding current, respectively, while η_2 is the arc thermal efficiency. a_1 , a_2 , b , and c are the geometric parameters of the double-ellipsoidal heat source model.

2.2. Molten pool driving-force model

2.2.1. Recoil pressure

During welding, when the heat-source energy density applied to the material surface exceeds a critical threshold, intense evaporation occurs, generating high-temperature metal vapor accompanied by a significant recoil force acting on the molten pool surface, thereby producing vapor recoil pressure. Under equilibrium conditions, the metal vapor pressure $\vec{P}$ can be described using the Clausius-Clapeyron equation [34]:

$$\frac{d\vec{P}}{dT} = \frac{\Delta H_v}{T_g \Delta V_v} \approx \frac{\Delta H_v}{T_g V_v} \quad (3)$$

Where T_g is the vaporization temperature of the substrate, and ΔH_v is the enthalpy of vaporization.

2.2.2. Arc pressure

The arc pressure acts directly on the molten pool surface through the plasma, and its spatial distribution is commonly represented by a Gaussian-type arc pressure function [35]:

$$P_a = \frac{c_j \mu_m I^2}{8\pi \sigma_j^2} \exp\left(\frac{-r_a^2}{2\sigma_j^2}\right) \quad (4)$$

Where μ_m is the vacuum magnetic permeability, r_a is the arc column radius, and σ_j is the arc heat flux distribution coefficient, which satisfies the following relationship with the welding current I :

$$\sigma_j = 0.533I^{0.294} \quad (5)$$

2.2.3. Electromagnetic force

During welding, the arc current generates a magnetic field in the surrounding space, causing co-directional current-carrying elements to attract each other under the magnetic force, thereby inducing an electromagnetic force within the molten pool. This force can be calculated using [Eq. \(6\)](#) [36,37]:

$$\begin{cases} F_{e-x} = -\frac{\mu_m I^2}{4\pi^2 \sigma_j^2 r_a} \exp\left(-\frac{r_a^2}{2\sigma_j^2}\right) \times \left[1 - \exp\left(-\frac{r_a^2}{2\sigma_j^2}\right)\right] \left(1 - \frac{z}{c_m}\right)^2 \frac{x}{r_a} \\ F_{e-y} = -\frac{\mu_m I^2}{4\pi^2 \sigma_j^2 r_a} \exp\left(-\frac{r_a^2}{2\sigma_j^2}\right) \times \left[1 - \exp\left(-\frac{r_a^2}{2\sigma_j^2}\right)\right] \left(1 - \frac{z}{c_m}\right)^2 \frac{y}{r_a} \\ F_{e-z} = -\frac{\mu_m I^2}{4\pi^2 r_a^2 c_m} \left[1 - \exp\left(-\frac{r_a^2}{2\sigma_j^2}\right)\right]^2 \left(1 - \frac{z}{c_m}\right) \end{cases} \quad (6)$$

Where c_m denotes the thickness of the aluminum alloy substrate.

2.2.4. Surface tension

During welding, instantaneous heating followed by rapid cooling generates a pronounced temperature gradient within the molten pool. This gradient induces variations in the surface tension P_s of the molten pool, which can be evaluated using [Eqs. \(7\)-\(9\)](#) [38,39]:

$$P_s = \gamma \kappa \quad (7)$$

$$\gamma = \gamma_0 + \frac{d\gamma}{dT}(T - T_0) \quad (8)$$

$$\kappa = - \left[\nabla \cdot \left(\frac{\mathbf{n}}{|\mathbf{n}|} \right) \right] = \frac{1}{|\mathbf{n}|} \left[\left(\frac{\mathbf{n}}{|\mathbf{n}|} \cdot \nabla \right) |\mathbf{n}| - (\nabla \cdot \mathbf{n}) \right] \quad (9)$$

Where γ is the surface tension coefficient, κ is the surface curvature, T_0 is the solidus temperature of the substrate, γ_0 is the surface tension coefficient at room temperature, and $\mathbf{n}$ is the surface normal vector.

For aluminum-alloy molten pools, the surface tension generally decreases with increasing temperature. When a pronounced temperature gradient exists on the molten pool surface, spatial variations in surface tension arise, which in turn drive the liquid metal to flow along the surface. This surface flow proceeds from regions of higher temperature toward regions of lower temperature, and the driving force is the Marangoni force induced by the surface-tension gradient. The Marangoni force can be calculated using Eq. (10) [39]:

$$\mathbf{F}_m = - \frac{d\gamma}{dT} \frac{dT}{ds} \quad (10)$$

Where T is the fluid temperature and $\mathbf{s}$ is the surface tangential vector.

2.2.5. Thermal buoyancy

In addition to altering the surface tension, the temperature gradient within the molten pool also induces variations in the density of the liquid metal, thereby generating buoyancy-driven convective flow. The corresponding buoyant force can be expressed as [40]:

$$\mathbf{F}_b = \rho \alpha g \Delta T \quad (11)$$

Where α is the thermal expansion coefficient of the material, and g is the gravitational acceleration.

2.3. Droplet transition model

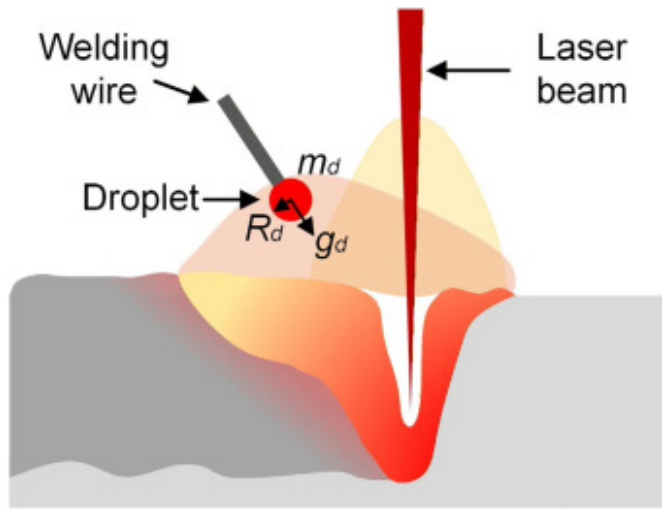
In laser-arc hybrid welding, droplet transfer exerts a significant influence on the thermo-fluid-mechanical behavior of the molten pool as well as on the metal filling characteristics. To account for the effects of droplet transfer, a spherical fluid element is introduced above the computational domain in this study, where the droplet is simplified as a liquid sphere with an initial radius

R_d , temperature T_d , and velocity v_d , as shown in Fig. 1. The corresponding mass, energy, and momentum source terms are coupled into the governing equations of molten-pool dynamics [41,42]. The droplet parameters are obtained from experimental observations: $T_d = 1500\text{ K}$, $R_d = 0.6\text{ mm}$, an initial position 2 mm above the molten pool, and a transfer frequency $f_d = 133\text{ Hz}$. The initial velocity v_d of the droplet is determined according to the mass conservation relationship:

$$m_w = \pi R_w^2 \rho_s v_f = m_s f_d = m_d \quad (12)$$

$$m_s = \frac{4}{3} \pi \rho_l R_d^3 \quad (13)$$

where m_w is the mass of the molten wire, m_d is the mass of the droplet, m_s represents the mass source of a single droplet, R_w is the wire radius, ρ_s and ρ_l are the densities of the wire and the droplet, respectively, and v_f is the wire feeding speed.



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Fig. 1. Droplet transition model.

Accordingly, the additional energy q_d transferred from the droplet to the molten pool per unit time and the impact force F_d exerted on the molten pool during droplet transfer can be further calculated as:

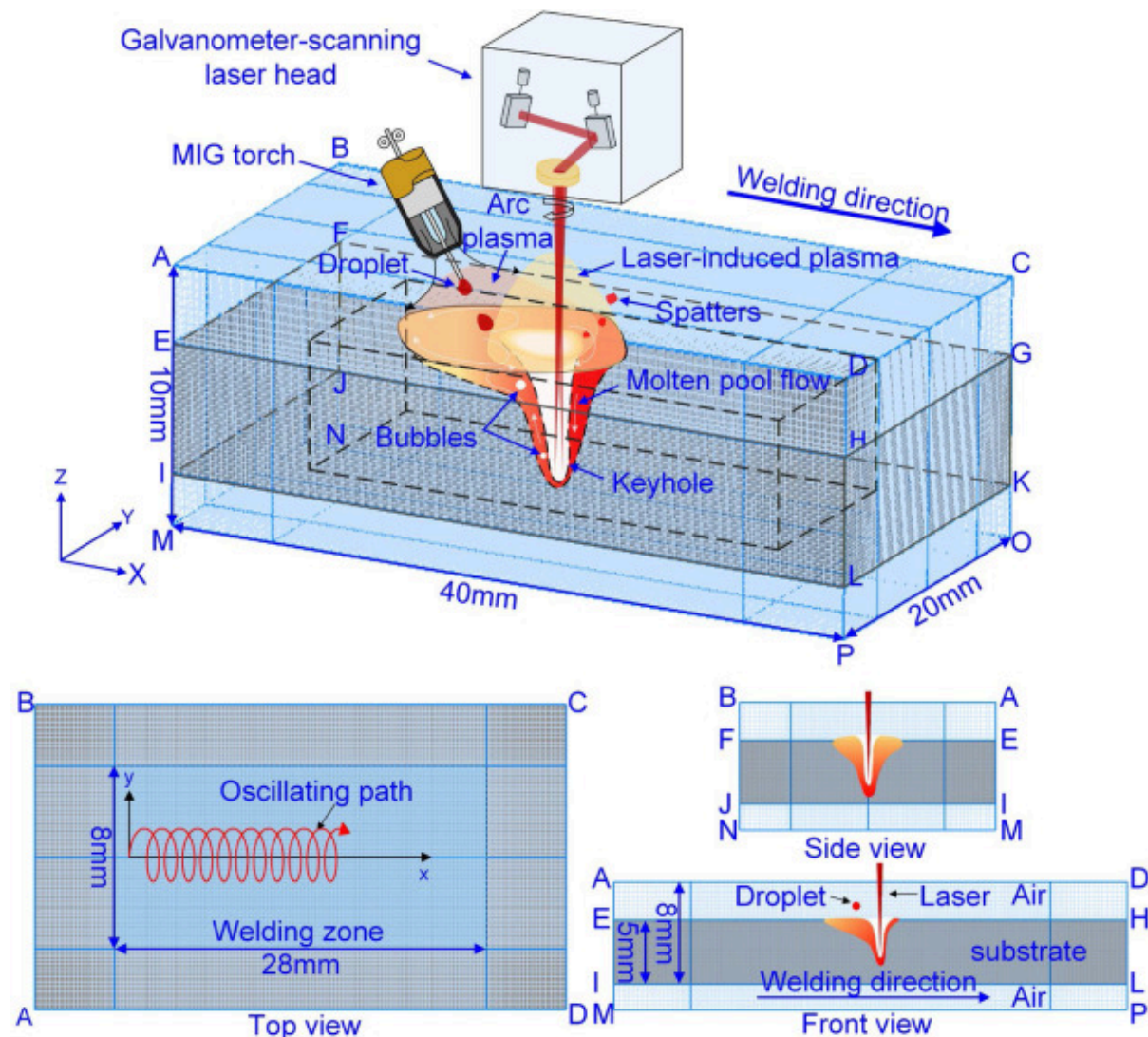
$$q_d = \frac{4}{3} C_d \pi \rho_l R_d^3 (T_d - T) \quad (15)$$

$$F_d = m_s g_d = \frac{4}{3} \pi \rho_l g_d R_d^3 \quad (16)$$

where g_d is the impact acceleration of the droplet driven by the arc force.

2.4. Mesh division and boundary conditions

In this study, the computational domain was defined as a cuboid with dimensions of 40 mm × 20 mm × 10 mm and discretized using a non-uniform mesh. As illustrated in Fig. 2, the domain was divided into three layers along the Z-direction: the upper and lower layers were gaseous regions with thicknesses of 3 mm and 2 mm, respectively, while the middle layer represented the aluminum alloy substrate with a thickness of 5 mm.



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Fig. 2. Geometric mesh model and dimensions.

The welding region was discretized using a linearly graded mesh, with the minimum grid resolution of 0.1 mm at the weld center, which gradually increased outward to ensure a smooth transition to the non-welding region. In the non-welding region,

the mesh size was gradually increased from 0.15 mm toward the coordinate boundaries up to 0.3 mm, thereby achieving a balance between computational accuracy and efficiency.

For the gaseous region, an air stream was introduced as the shielding gas. Its density, temperature, and pressure parameters were defined according to the physical properties of nitrogen used in the experiments, and they satisfy the ideal gas law [43]:

$$pV = nRT \quad (17)$$

where p and V is the gas pressure and volume, respectively, and R is the gas constant.

For the substrate, AA6061 aluminum alloy was used as the base material, and its thermophysical properties are listed in Table 1.

Table 1. Thermal-physical properties of AA6061 aluminum alloy.

Thermophysical properties	symbol	units	value
Solidus temperature	T_s	K	875
Liquidus temperature	T_l	K	924
Boiling temperature	T_g	K	2790
Convective heat transfer coefficient	h_c	$\text{W m}^{-1}\text{K}^{-1}$	35
Stefan-Boltzmann constant	σ	$\text{W m}^{-2}\text{K}^{-4}$	5.67×10^8
Emissivity	ε	-	0.9
Surface tension coefficient	γ_0	N m^{-1}	0.88
Temperature coefficient of surface tension	$d\gamma/dT$	$\text{N m}^{-1} \text{K}^{-1}$	-1.35×10^{-4}
Coefficient of thermal expansion	α	K^{-1}	9.18×10^{-5}
Latent heat of fusion	H_{sl}	J kg^{-1}	5.03×10^5
Latent heat of the vapor	ΔH_v	J kg^{-1}	1.07×10^7

Thermophysical properties	symbol	units	value
Ambient temperature	T_0	K	300

To ensure the accuracy and physical reliability of the computational results, the boundary conditions must be appropriately defined. The boundary conditions of the computational domain mainly include energy boundaries and momentum boundaries. The energy boundary describes the balance between heat-source input and heat transfer within the molten pool and is expressed by Eq. (18) [44,45]. The momentum boundary characterizes the momentum balance on the keyhole wall and is given by Eq. (19) [26].

$$q_1(x, y, z) + q_2(x, y, z) + q_d - k \frac{\partial T}{\partial n} = q_c + q_r + q_{vap} \quad (18)$$

$$-P_l + 2\mu \frac{\partial V_n}{\partial n} = -P_r + P_a + P_d + P_s \quad (19)$$

Where q_1 is the energy input from the laser heat source, q_2 is the energy input from the arc heat source, and q_d is the heat carried by the droplets. q_c , q_r , and q_{vap} denote the heat losses due to convection, radiation, and evaporation, respectively. P_l is the hydrostatic pressure, μ is the dynamic viscosity, V_n is the normal velocity component, P_r is the vapor recoil pressure, P_a is the arc pressure, P_d is the droplet impact pressure, and P_s is the surface tension.

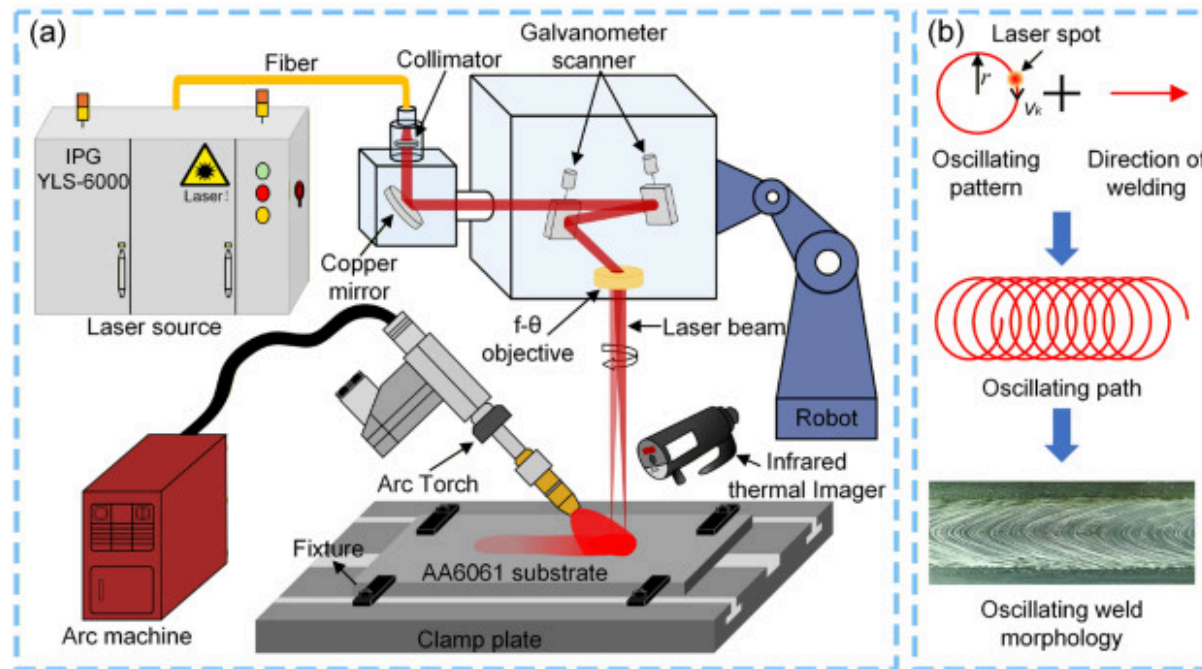
2.5. Experimental equipment and materials

As shown in Fig. 3a, the oscillating laser-arc hybrid welding experimental platform primarily consists of an IPG YLS-6000 fiber laser, a welding head equipped with a Scanlab galvanometer, a Fronius arc welding power source, and a Fanuc industrial robot. The base material used in the experiments is AA6061-T6 aluminum alloy plate with dimensions of 200 mm × 100 mm × 5 mm, while the filler material is ER5087 welding wire with a diameter of 1.2 mm. Prior to welding, the substrate was polished with an abrasive wheel and cleaned with acetone to remove surface oxide films and oil contaminants. During welding, the laser beam oscillated in a circular oscillating mode (Fig. 3b, the laser beam oscillating velocity v_k can be approximated by the average travel speed $\bar{v}_k$, as given in Eq. (20)), guiding the arc to achieve flat position deposition. The laser focal radius was set to 0.25 mm, with a defocus distance of 0 mm. Throughout the process, argon gas with a purity of 99.99 % was employed as the shielding gas. The

main process parameters are summarized in Table 2.

$$v_k = \bar{v}_k = 2\pi r f \quad (20)$$

Where f is the laser beam oscillating frequency and r is the oscillating radius.



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Fig. 3. Schematic of the oscillating laser-arc hybrid welding experimental system.

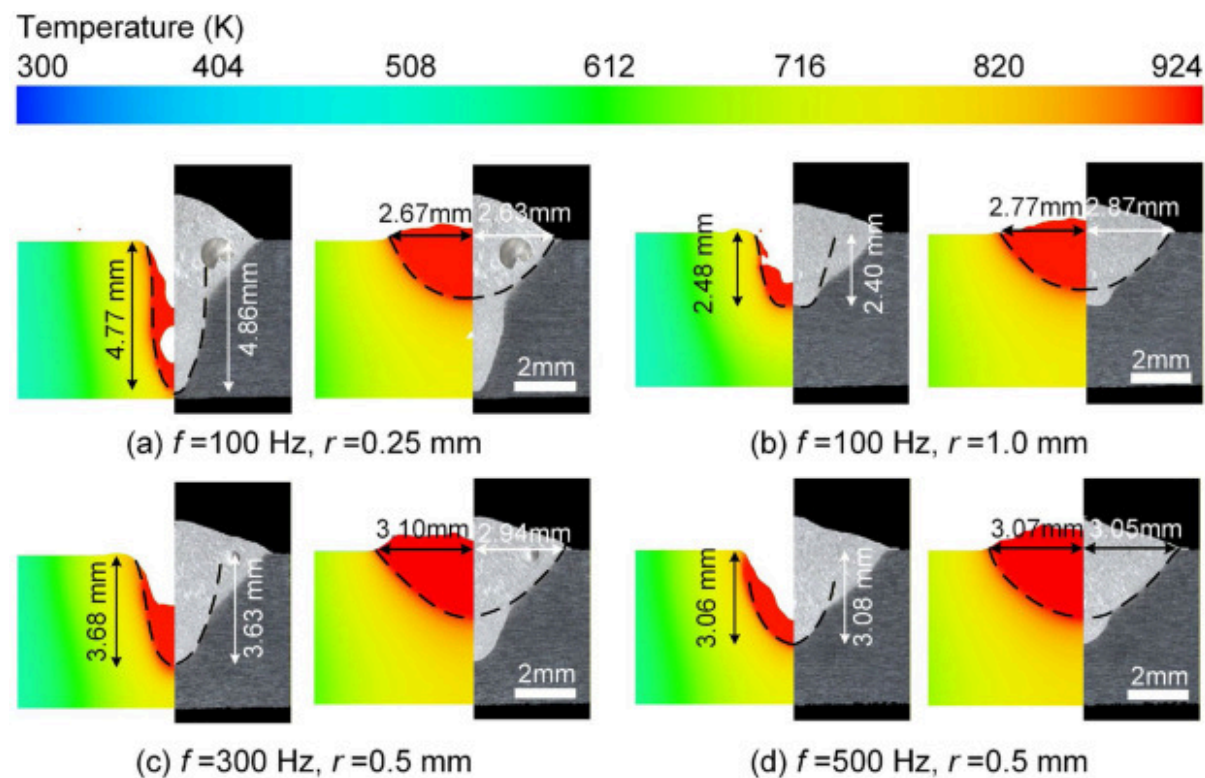
Table 2. Welding process parameters.

Laser power P (kW)	Arc current I (A)	Welding speed v_w (m/min)	Laser-to-Wire distance d_{LA} (mm)	Oscillating radius r (mm)	Oscillating frequency f (Hz)
4.0	130	2.0	2.0	0~1	0~500

3. Results and discussion

3.1. Model validation

To ensure the accuracy and reliability of the developed numerical model, validation was conducted from two aspects: weld geometry and molten pool thermo-fluid behavior. With respect to weld geometry, since the locations of the maximum penetration depth and the maximum weld width do not lie on the same cross-section, the laser-side section and the arc-side section were selected separately for comparative analysis. [Fig. 4.](#) presents the comparison between the numerical predictions and the experimentally observed weld cross-sections, and the corresponding quantitative results are summarized in [Table 3.](#) As shown in the [Fig. 4.](#) and the table, under identical processing parameters, the simulated weld profiles and dimensions exhibit good agreement with the experimental measurements. The maximum deviation in weld penetration depth across all oscillating parameters is below 4 %, indicating that the established model can accurately capture the formation characteristics of oscillating laser-arc hybrid welding.



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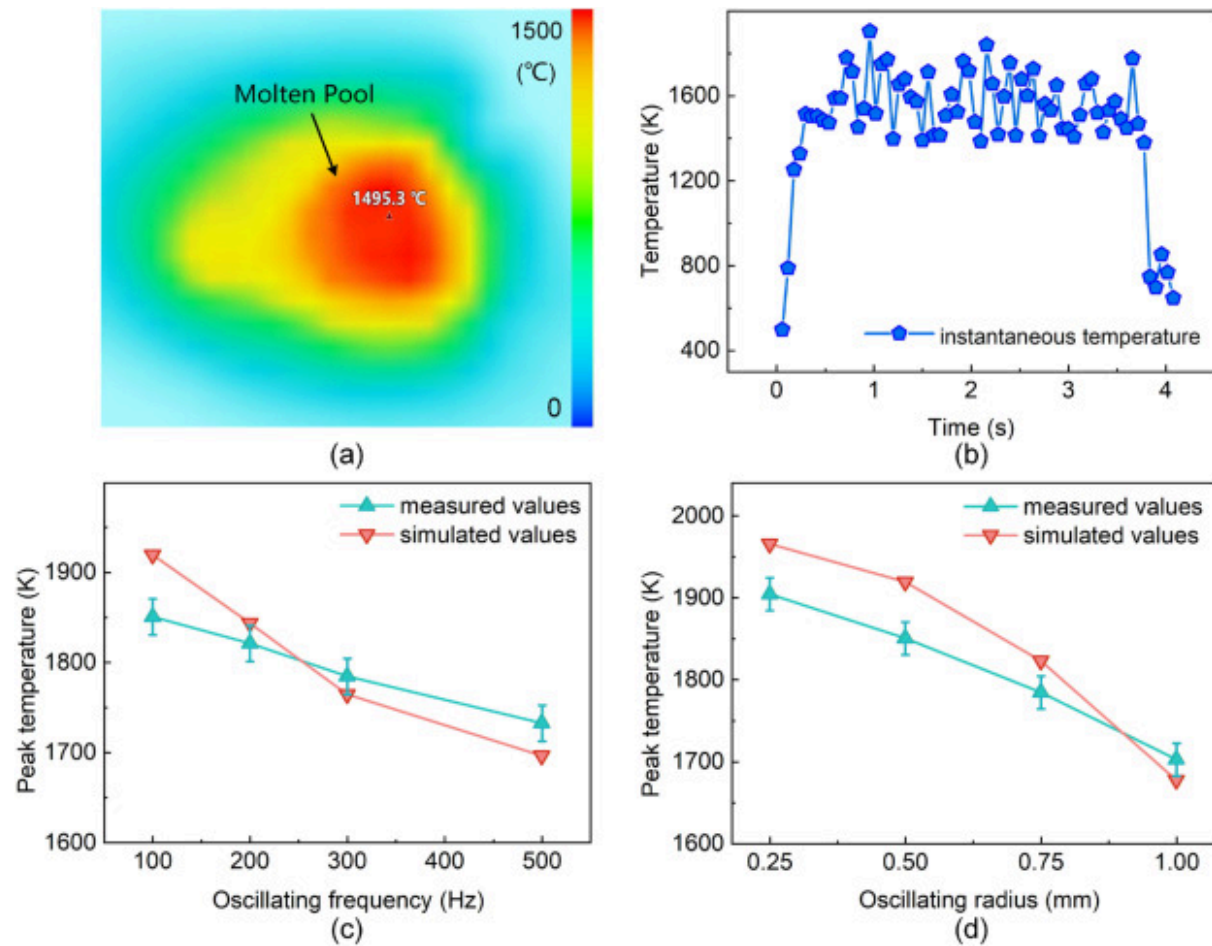
Fig. 4. Comparison between simulated and experimental weld cross-sections: (a) $f=100$ Hz, $r=0.25$ mm; (b) $f=100$ Hz, $r=1.0$ mm; (c) $f=300$ Hz, $r=0.5$ mm; (d) $f=500$ Hz, $r=0.5$ mm.

Table 3. Dimensions of the calculated and experimental weld.

Oscillating parameters	Penetration depth			Width		
	Calculated	Measured	Precision	Calculated	Measured	Precision
$f=100$ Hz, $r=0.25$ mm	4.77	4.86	98.1 %	2.67	2.63	98.5 %
$f=100$ Hz, $r=1.0$ mm	2.48	2.40	96.8 %	2.77	2.87	96.5 %

Oscillating parameters	Penetration depth			Width		
	Calculated	Measured	Precision	Calculated	Measured	Precision
$f=300$ Hz, $r=0.5$ mm	3.68	3.63	98.6 %	3.10	2.94	94.8 %
$f=500$ Hz, $r=0.5$ mm	3.06	3.08	99.4 %	3.07	3.05	99.3 %

In terms of molten pool thermo-fluid behavior, a FOTRIC 348x+ intelligent infrared thermal imager was employed to acquire thermal images of the molten pool, and the surface peak temperature data under different oscillating parameters were extracted using AnalyzIR software for validation of the numerical simulation results. As shown in Fig. 5a, after aligning the thermal imager with the centerline of the welding path, the surface temperature of the molten pool can be measured and recorded in real time during the welding process. The recorded temperature fields were subsequently visualized as color thermal images using the AnalyzIR software.



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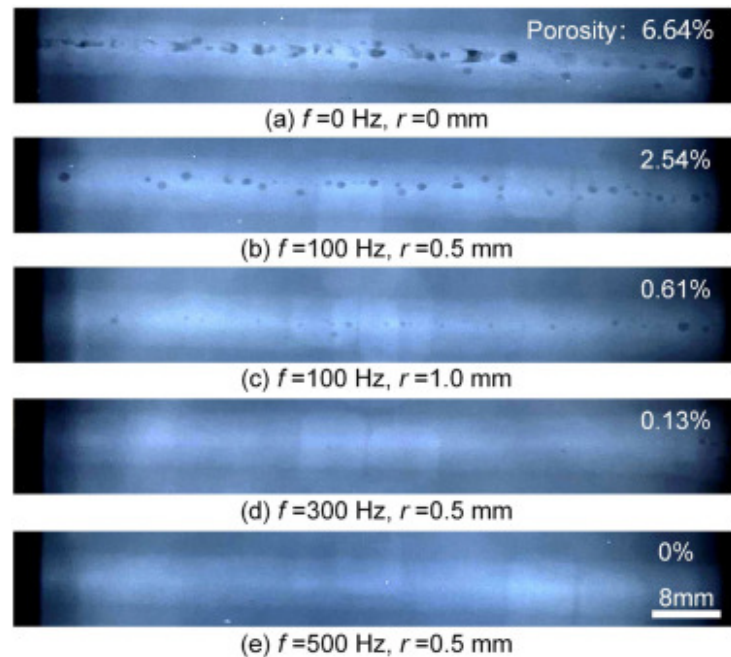
Fig. 5. Measurement of molten pool temperature during welding: (a) thermal imaging of the molten pool; (b) instantaneous temperature of the molten pool ($f = 100$ Hz, $r = 0.5$ mm); (c) variation of peak temperature with oscillating frequency ($r = 0.5$ mm); (d) variation of peak temperature with oscillating radius ($f = 100$ Hz).

Fig. 5b presents the instantaneous evolution of the surface temperature at the center of the molten pool during oscillating laser-arc hybrid welding. It can be observed that the transient surface temperature of the molten pool increases rapidly after the initiation of welding, reaches a quasi-steady oscillatory state after approximately 300 ms, and exhibits a temperature fluctuation

amplitude of up to 500 K, which persists until the end of the welding process. Fig. 5c and d illustrate the variations in the surface peak temperature of the molten pool as a function of laser beam oscillating frequency and oscillating radius, respectively. Both the experimental measurements and numerical simulations consistently indicate that the surface peak temperature of the molten pool gradually decreases with increasing oscillating frequency or oscillating radius. Specifically, when the oscillating frequency f increases from 100 Hz to 500 Hz, the measured surface peak temperature of the molten pool decreases from 1850.6 K to 1732.5 K, corresponding to a reduction of 6.4 %. Similarly, when the oscillating radius increases from 0.25 mm to 1.0 mm, the measured surface peak temperature decreases from 1904.1 K to 1702.3 K, representing a reduction of 10.6 %. In addition, comparison between the measured and simulated surface peak temperatures shows good agreement, with discrepancies below 100 K for all oscillating conditions. This phenomenon is consistent with the findings of Mohan et al. [24], who reported that laser beam oscillation can enhance molten pool stirring, thereby achieving a more uniform temperature distribution within the molten pool. These results demonstrate that the established thermo-fluid coupled model can accurately predict the temperature field distribution in the molten pool, providing a reliable theoretical basis for optimizing weld formation quality.

3.2. Effects of laser beam oscillating parameters on weld porosity

Fig. 6. shows the distribution characteristics of internal pores in welds produced under different oscillating parameters. Under non-oscillating conditions ($f = 0$ Hz, $r = 0$ mm), a large number of chain-like pores with irregular shapes and varying sizes are generated within the weld, mainly concentrated near the weld center, resulting in a high porosity of 6.64 %. After introducing laser beam oscillation, the number of chain-like pores is significantly reduced as the oscillating radius r increases from 0 mm to 1.0 mm, and the porosity decreases from 6.64 % to 0.61 %, corresponding to a reduction of 90.8 %. A similar pore-suppression effect is observed when the oscillating frequency f increases from 100 Hz to 500 Hz; in particular, when $f = 500$ Hz, the weld becomes completely pore-free. These results clearly demonstrate that increasing either the oscillating radius or frequency effectively suppresses pore formation, and dense, pore-free welds of high quality can be more readily achieved under high-frequency and large-radius oscillating conditions.



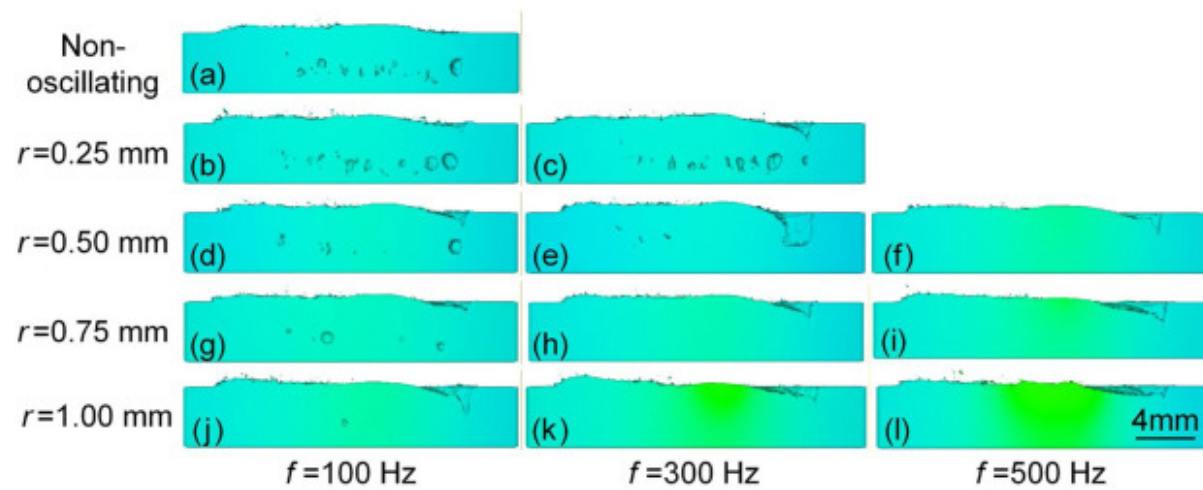
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Fig. 6. Porosity distribution and porosity rate in aluminum alloy welds: (a) $f = 0$ Hz, $r = 0$ mm; (b) $f = 100$ Hz, $r = 0.5$ mm; (c) $f = 100$ Hz, $r = 1.0$ mm; (d) $f = 300$ Hz, $r = 0.5$ mm; (e) $f = 500$ Hz, $r = 0.5$ mm.

Fig. 7. presents the distribution characteristics of simulated weld-section porosity under different oscillating parameters. Similarly, under non-oscillating conditions, a large number of chain-like pores form at the bottom of the weld, primarily concentrated near the fusion line. After introducing laser beam oscillation, increasing either the oscillating frequency f or oscillating radius r effectively reduces pore defects. Specifically, when $f \geq 300$ Hz and $r \geq 0.75$ mm, or $f \geq 500$ Hz and $r \geq 0.5$ mm, internal pores can be completely eliminated. This trend is in good agreement with the experimental observations shown in Fig. 6. Based on this, pore features within the simulated welds were accurately identified and quantified using a mesh-based feature extraction method, the result as illustrated in Fig. 8. It can be seen that increasing the oscillating frequency or radius maintains the weld porosity at a low level; In particular, when the oscillating parameters satisfy $f \geq 300$ Hz and $r \geq 0.5$ mm, the porosity can be reduced to below 0.5 %. Moreover, the numerical predictions exhibit a high degree of consistency with the experimental

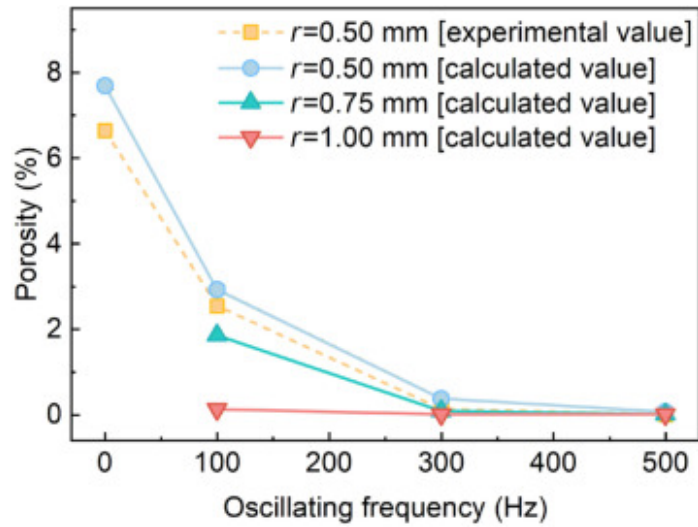
trends, with comparable values; under non-oscillating conditions, the maximum deviation between simulation and experiment is only 1.05 %.



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Fig. 7. Effect of laser beam oscillating frequency f and radius r on porosity.



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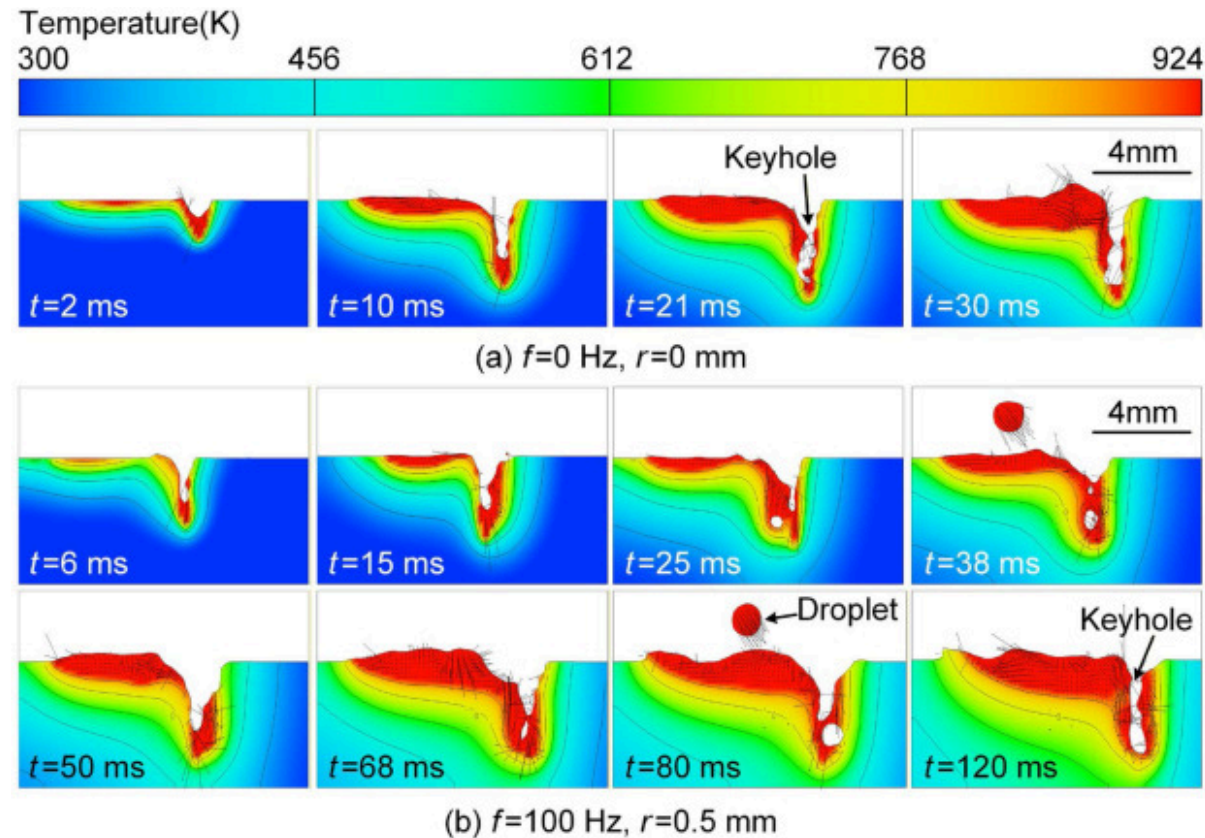
Fig. 8. Variation of weld porosity with oscillating frequency and oscillating radius.

Further comparison of the present study results with the existing literature reveals that the trend of weld porosity variation with oscillating parameters observed in this work is consistent with the patterns reported by Fetzer et al. [20] in oscillating laser welding. However, notable differences exist: in Fetzer et al.'s study, a pore-free weld was achieved at an oscillating frequency of only 200 Hz with an oscillating radius of 0.75 mm. In contrast, in the present study, weld porosity was not completely eliminated when the oscillating frequency was 300 Hz and the oscillating radius was 0.5 mm; a pore-free weld was only obtained when the oscillating frequency was further increased to 500 Hz. This discrepancy indicates that, compared to pure oscillating laser welding, the thermo-fluid behavior within the molten pool during oscillating laser-arc hybrid welding is more complex. This complexity primarily arises from the interaction between the laser and arc, droplet transition dynamics, and the simultaneous presence of strong vortex flows, all of which collectively influence keyhole stability, bubble formation, and their subsequent evolution.

3.3. Dynamic mechanisms of keyhole and porosity formation

3.3.1. Dynamic mechanisms of keyhole formation

During the laser-arc hybrid welding (LAHW) process, the behavior of the keyhole plays a crucial role in the formation of porosity. To elucidate the dynamic mechanism underlying this phenomenon, the evolution of the keyhole under both non-oscillating and circular oscillating conditions was comparatively analyzed, as shown in Fig. 9.



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Fig. 9. Dynamic evolution of the keyhole: (a) $f = 0$ Hz, $r = 0$ mm; (b) $f = 100$ Hz, $r = 0.5$ mm.

Under non-oscillating conditions (Fig. 9a), when the high-energy-density laser irradiates the material surface, the local metal rapidly melts and violently evaporates, generating a large amount of metallic vapor. The escaping vapor exerts a significant recoil

pressure on the keyhole wall, driving the molten metal downward and causing the keyhole to deepen progressively. Meanwhile, the keyhole continuously absorbs laser energy and converts it into thermal energy, further promoting melting and evaporation of the surrounding metal. This process leads to continuous vapor accumulation and pressure buildup inside the keyhole, eventually forming a downward-depressed deep keyhole. When the vapor pressure reaches equilibrium with the hydrostatic pressure and surface tension of the molten metal, the keyhole depth stabilizes, marking the onset of a steady-state stage. This process occurs rapidly, taking approximately 30 ms from laser incidence to keyhole stabilization. Fig. 9b shows the keyhole evolution under circular oscillating conditions. The fundamental formation mechanism remains similar to that under non-oscillating conditions, with metal vapor pressure acting as the dominant driving force. However, compared with the non-oscillating case, the time required for the keyhole to reach steady state is significantly prolonged, being approximately three to four times longer. This can be attributed to two primary factors: (1) During laser oscillating, the energy input varies dynamically, causing repeated remelting of the keyhole region and resulting in a nonlinear evolution of keyhole depth; (2) The oscillating motion enhances heat transfer within the molten pool, making it difficult for heat to accumulate rapidly inside the keyhole, thereby slowing the buildup of metal vapor pressure and delaying steady-state formation.

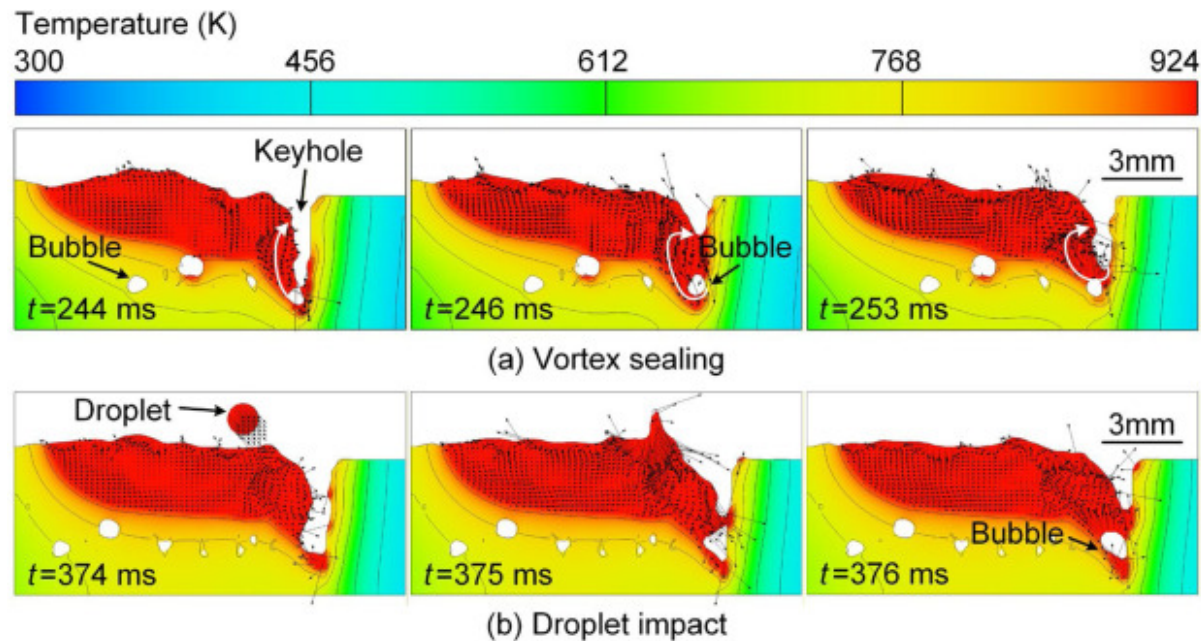
Overall, the fundamental mechanism underlying keyhole formation lies in the fact that a high-energy-density laser induces localized high-temperature and high-pressure regions within the molten pool, causing the liquid metal to flow downward under the strong pressure of metal vapor, thereby forming a keyhole. The stability of the keyhole depends on the dynamic balance among the metal vapor pressure, hydrostatic pressure, and surface tension. Only when these three forces reach equilibrium simultaneously can the keyhole be stably established and maintained in a steady state.

3.3.2. Dynamic mechanisms of pore formation

For keyhole-type pores, their formation mechanism primarily originates from the instability and collapse of the keyhole, which can be categorized into two types: direct closure induced by vortex effects and indirect closure caused by droplet impact.

Fig. 10a illustrates the keyhole closure process induced by the vortex effect in laser-arc hybrid welding. In this process, the recoil pressure of metal vapor drives the molten metal at the keyhole bottom to form a clockwise vortex. Under the action of this vortex, the molten metal at the keyhole shoulder contacts the front wall and forms a liquid bridge, thereby sealing the keyhole bottom and generating a bubble. Subsequently, the bubble is transported toward the rear region of the molten pool along the

vortex flow. Upon reaching the low-temperature thermal boundary, the bubble fails to escape in time and is ultimately captured by the solidification front, resulting in the formation of a pore.



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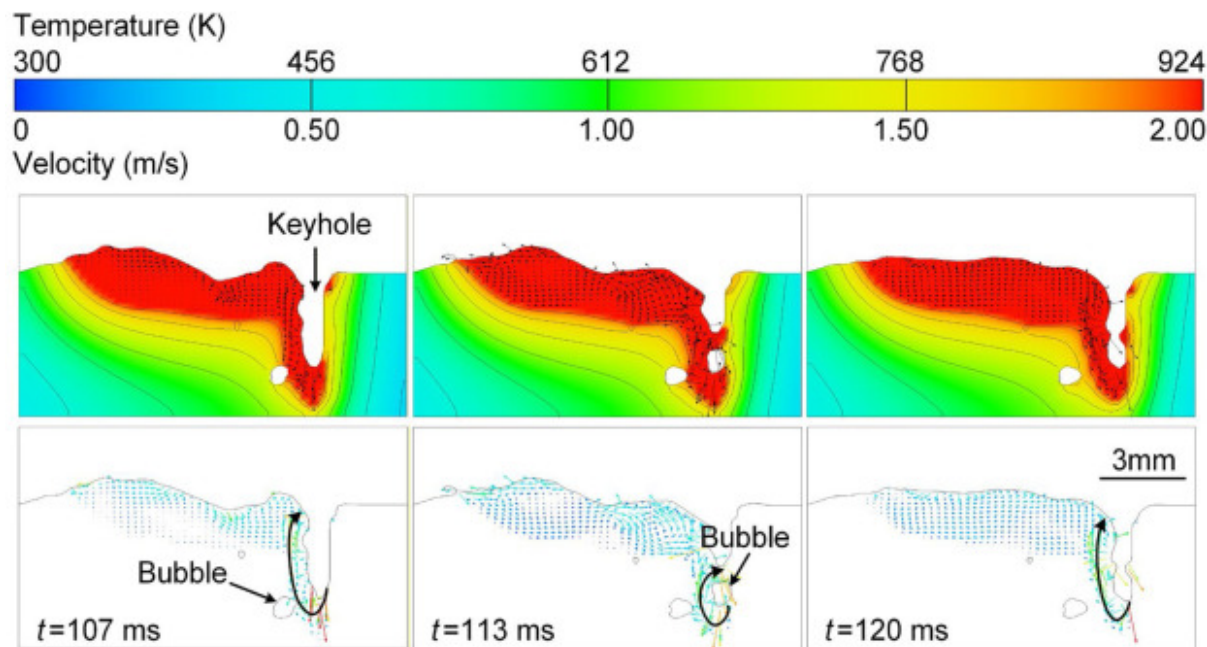
Fig. 10. Modes of porosity formation: (a) vortex sealing; (b) droplet impact.

Fig. 10b shows the indirect keyhole-closure process induced by droplet impingement in laser-arc hybrid welding. As illustrated, when a droplet transfers into the molten pool, its intense impact momentum drives the molten metal to flow forward and compresses the keyhole, causing the root of the keyhole to close and generating a bubble. This bubble is likewise transported toward the rear of the molten pool under the vortex flow. However, because it fails to escape in time, it is eventually captured by the solidification front and transformed into a pore.

The above dual mechanisms—vortex-driven and droplet-impact-induced bubble formation—highlight the combined influence of molten pool flow dynamics and droplet transfer on keyhole stability and pore generation. Among these mechanisms, the vortex-driven porosity formation observed in this study is, to some extent, consistent with the results reported by Huang et al. [10], who

demonstrated that a clockwise vortex at the bottom of the molten pool, driven by recoil pressure, can promote keyhole collapse and closure, thereby leading to bubble generation. This indicates that vortex effects play a critical role in porosity formation. However, unlike the present study, Huang et al. only investigated vortex-driven bubble formation and did not consider the impact of droplet transfer. The results of the current study demonstrate that droplet transfer can also affect keyhole stability and further promote bubble formation.

Fig. 11. illustrates the complete evolution process of pore formation in the molten pool during laser-arc hybrid welding under non-oscillating conditions. As shown, when the keyhole remains open, the laser beam energy directly reaches the bottom of the keyhole, leading to a gradual increase in metal vapor recoil pressure and Marangoni force. These forces drive the molten metal to form a clockwise vortex in the rear region of the keyhole. Subsequently, the vortex pushes the molten metal located at the mid-rear region of the keyhole to flow forward and come into contact with the front wall of the keyhole, thereby forming a liquid bridge. This interaction induces keyhole instability and closure, resulting in the generation of a bubble. During this process, both the keyhole depth and the metal vapor recoil pressure decrease accordingly.



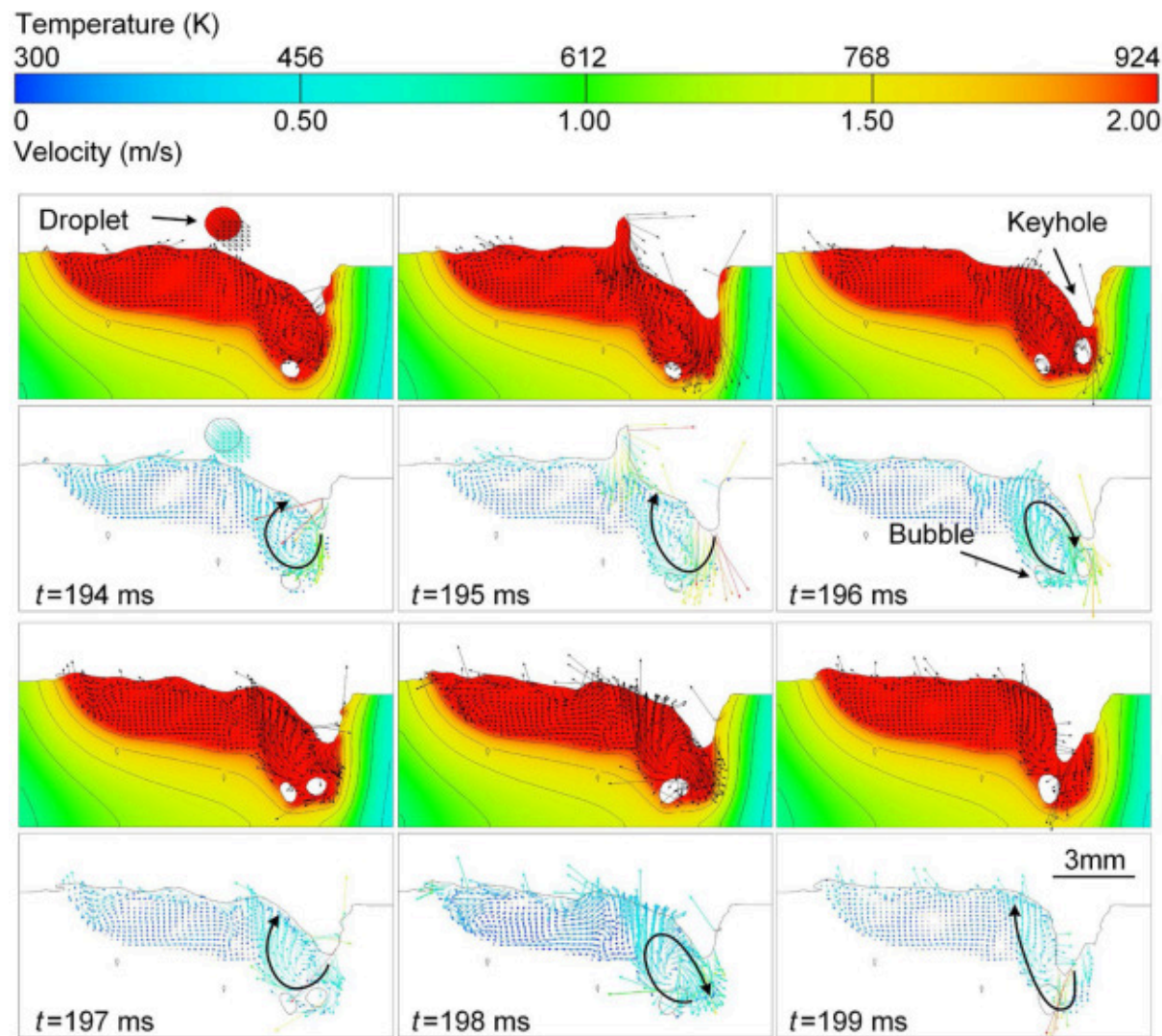
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Fig. 11. Temperature and flow fields during porosity formation in the molten pool under non-oscillating conditions.

However, this keyhole closure behavior is characterized by a pronounced transient and intermittent nature. When the laser energy is re-concentrated at the keyhole and the metal vapor recoil pressure increases again, the keyhole reopens. During this repeated cyclic process, the generated bubble gradually escapes from the laser interaction region and migrates toward the rear of the keyhole under the action of the vortex flow.

By comparison, the bubble formation process in the molten pool during oscillating laser-arc hybrid welding is more complex, as shown in Fig. 12, at $t = 194$ ms, a droplet is located above the molten pool, while a bubble with a diameter of approximately 0.6 mm has already formed at the bottom of the pool and migrates rearward at a velocity of 0.62 m/s under the action of the vortex flow. Subsequently, the momentum transferred by the impinging droplet accelerates the clockwise vortex toward the front, compressing and closing the keyhole and generating a new bubble with a diameter of approximately 0.8 mm. Driven by the metal vapor recoil pressure, both bubbles are carried toward the rear of the keyhole by the vortex flow. Because the newly formed bubble is closer to the keyhole and experiences a stronger vortex effect, it migrates at a higher velocity of 1.08 m/s, which is faster than the previously generated bubble. As a result, the two bubbles gradually approach each other and eventually merge into a larger bubble with a diameter of approximately 1.2 mm. During subsequent solidification, this bubble fails to escape and is ultimately trapped by the solid-liquid interface, evolving into a pore. These observations demonstrate that both the vortex effect and droplet impingement exert significant influence on bubble generation, migration, and interaction.



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Fig. 12. Temperature and flow fields during porosity formation in a circular oscillating molten pool($f = 100$ Hz, $r = 0.5$ mm).

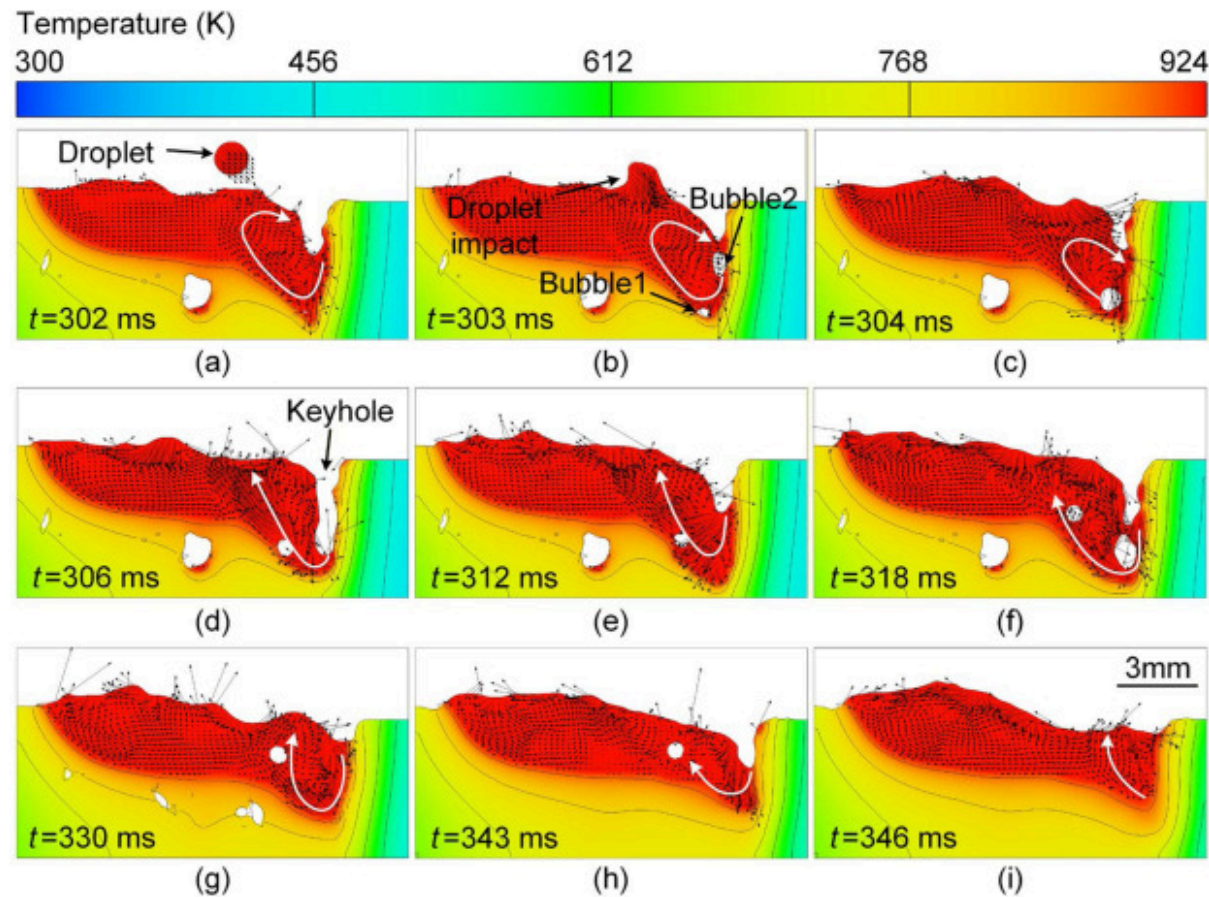
It should be noted that, whether under non-oscillating or oscillating conditions, the vortex effect and droplet impact effect are respectively dominated by the laser heat source and the arc heat source. Due to the interactive coupling between these two heat

sources, porosity formation is typically the result of the combined action of both effects.

3.4. Dynamics of porosity suppression induced by laser beam oscillation

3.4.1. Beam oscillation accelerates bubble escape from the molten pool

[Fig. 13](#). illustrates the entire process of bubble formation and escape from the molten pool under beam oscillating conditions. As shown in [Fig. 13a, b and c](#), at $t = 303$ ms, droplet impingement destabilizes and closes the keyhole, resulting in bubble formation. Subsequently, under the action of the vortex flow, the two bubbles located behind the keyhole gradually coalesce. During the forward oscillating of the laser beam, the coalesced bubble continues to migrate rearward with the vortex and rise upward ([Fig. 13d, e and f](#)). At $t = 318$ ms, the bubble successfully passes across the thermal-flow boundary at the bottom of the arc region, indicating that it will not be captured by the solidification front and evolve into a pore. Thereafter, at $t = 330$ ms, the bubble rises to a position approximately 1 mm below the molten pool surface. However, due to the large surface tension of the high-temperature molten metal near the pool surface, the bubble experiences a substantial resistance to upward motion and thus remains nearly stationary for the next two oscillating cycles. It finally escapes from the molten pool at $t = 346$ ms.



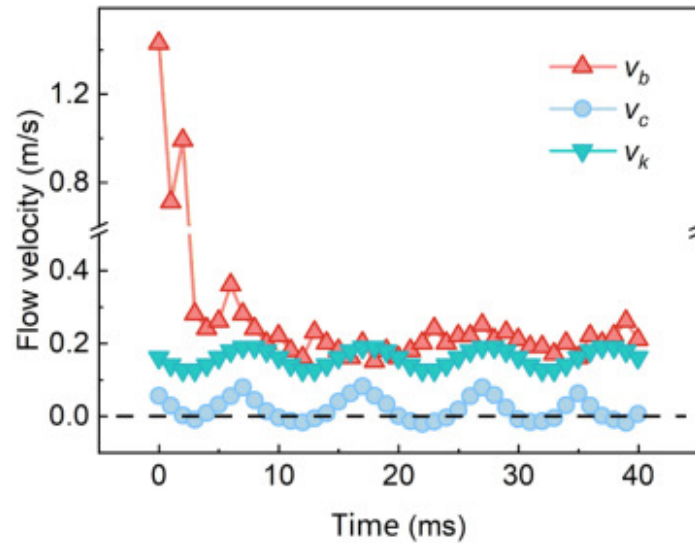
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Fig. 13. Evolution of bubbles from formation to escape from the molten pool ($f = 100$ Hz, $r = 0.25$ mm): (a-c) bubble formation and coalescence; (d-f) bubble migration and buoyant rise; (g-i) bubble escape from the molten pool.

Fig. 14. further compares the bubble migration velocity v_b , molten-pool solidification velocity v_c , and laser oscillating velocity v_k during the bubble ascent process. It can be observed that, after bubble formation, the vortex-driven momentum propels the bubble to a maximum migration velocity of 1.43 m/s, which gradually decreases during ascent and eventually stabilizes at approximately 0.2 m/s. The keyhole, guided by the oscillating laser beam, undergoes circular motion along the oscillating path, with its instantaneous velocity exhibiting periodic fluctuations: a period of 10 ms, a baseline velocity of 0.126 m/s, and a peak

velocity of 0.19 m/s. In contrast, the molten pool solidification velocity is significantly lower than both the bubble migration velocity and the laser oscillating velocity. Influenced by the backward remelting effect of the laser beam, the solid-liquid interface moves rearward during the latter half of each oscillating cycle, resulting in v_c periodically fluctuating within the range of -0.018 to 0.089 m/s, with a period matching that of the laser beam oscillation.



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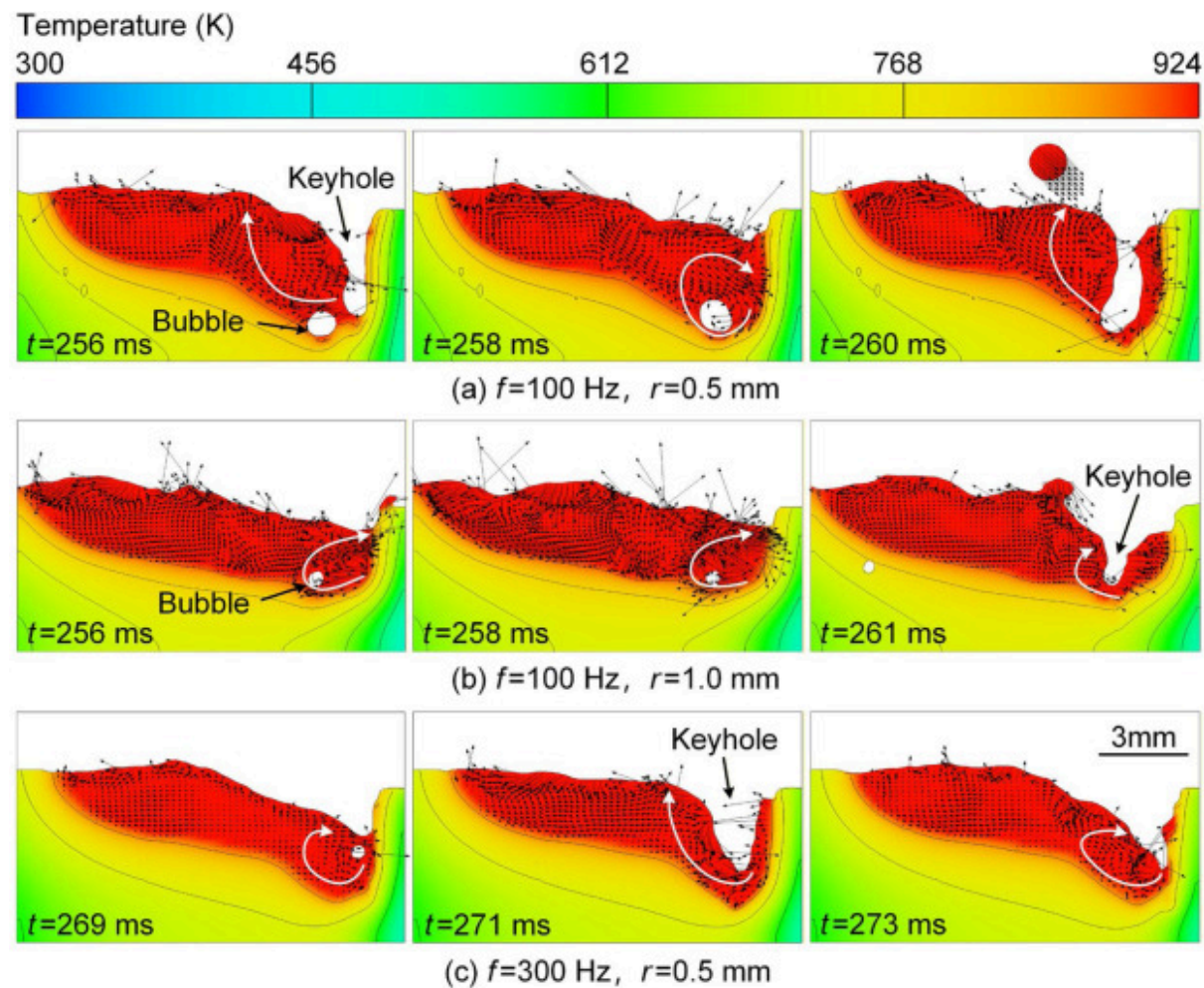
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Fig. 14. Velocity variations during bubble escape from the molten pool ($f = 100$ Hz, $r = 0.25$ mm).

It should be noted that a bubble migration velocity higher than the molten pool solidification velocity does not imply that the bubble will not be captured by the solidification front and form a pore. Whether the bubble escapes the molten pool depends on whether its required escape time is shorter than the pool solidification time [46]. Furthermore, since the bubble migration velocity exceeds the oscillating velocity of the laser beam, the likelihood of bubble interaction with the laser keyhole during the aforementioned process is relatively low. Under this condition, the laser primarily enhances the molten pool flow through oscillating-induced stirring, thereby accelerating bubble ascent and escape and reducing the likelihood of pore formation in the weld. This mode can be regarded as the first type of porosity suppression mechanism.

3.4.2. Bubble merging and absorption induced by laser keyhole rotation

Fig. 15. illustrates the coalescence and absorption of bubbles by the keyhole under different oscillating parameters. As shown in Fig. 15a, when the oscillating frequency is $f = 100$ Hz and the oscillating radius is $r = 0.5$ mm, the keyhole exhibits a relatively high depth-to-width ratio, making its base prone to rupture under intense hydrostatic pressure, thereby generating a bubble. During subsequent migration, this bubble is captured by the bottom boundary of the molten pool and loses its upward driving force. However, as the laser beam rotates backward, the keyhole re-encounters the bubble and achieves coalescence and absorption, partially suppressing pore formation. Furthermore, when the oscillating radius is increased to $r = 1$ mm (Fig. 15b), although the risk of keyhole base closure and bubble formation remains, the larger oscillating radius expands the motion range of the keyhole, enhancing its ability to interact with bubbles over a wider area and thereby absorb them more effectively. When the oscillating frequency is increased to $f = 300$ Hz while maintaining $r = 0.5$ mm (Fig. 15c), the laser beam exhibits a higher oscillating velocity, enabling the keyhole to rapidly encounter and coalesce with the bubble after its formation, thereby effectively suppressing pore formation. This phenomenon is consistent with the observations of Liu et al. [27] in narrow-gap oscillating laser–wire welding of aluminum alloys, where the periodic motion of the keyhole can actively capture bubbles in the molten pool. More importantly, the present study further confirms that this mechanism also operates under the more complex conditions of oscillating laser–arc hybrid welding, indicating that the porosity-suppression mechanism has a certain universality in beam-oscillated welding and further advances the theoretical understanding of porosity formation and suppression in oscillating laser–arc hybrid welding. In summary, under the guidance of the laser beam, the keyhole can actively capture and absorb bubbles through rotational motion, effectively inhibiting pore formation. Therefore, this approach can be regarded as the second type of porosity-suppression mechanism.



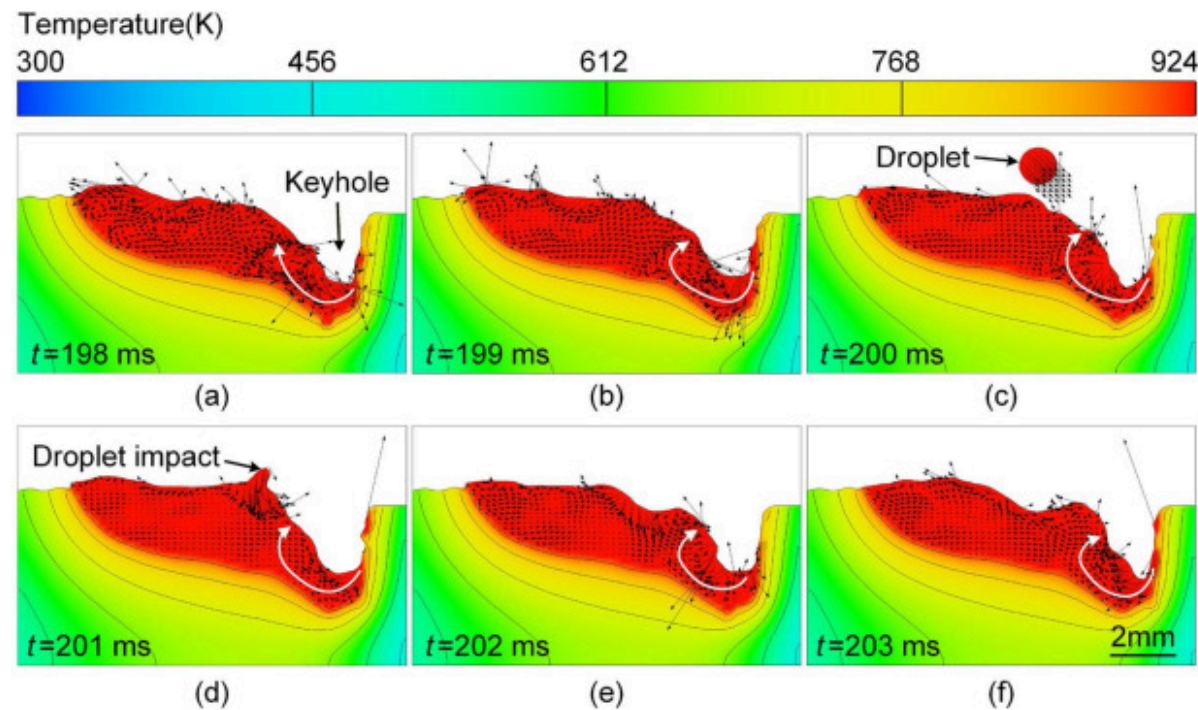
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Fig. 15. Bubble coalescence and absorption by the laser keyhole: (a) $f = 100$ Hz, $r = 0.5$ mm; (b) $f = 100$ Hz, $r = 1.0$ mm; (c) $f = 300$ Hz, $r = 0.5$ mm.

3.4.3. Beam oscillation enhance keyholes stability and suppresses porosity

Fig. 16. illustrates the dynamic evolution and stability characteristics of keyholes under high-frequency oscillating conditions. When the oscillating frequency is $f = 500$ Hz, the laser beam completes a oscillating cycle in 2 ms. During the time interval $t = 198$ – 203 ms, the laser beam undergoes three oscillating cycles accompanied by a single droplet transition. In this process, the centrifugal effect induced by the oscillating motion causes the keyhole cross-section to adopt a “wide and shallow” morphology, with a relatively flat bottom, and maintains a high level of stability throughout the oscillating cycles. Even after droplet impact on the molten pool, the keyhole remains open and does not close, thereby fundamentally suppressing pore formation.



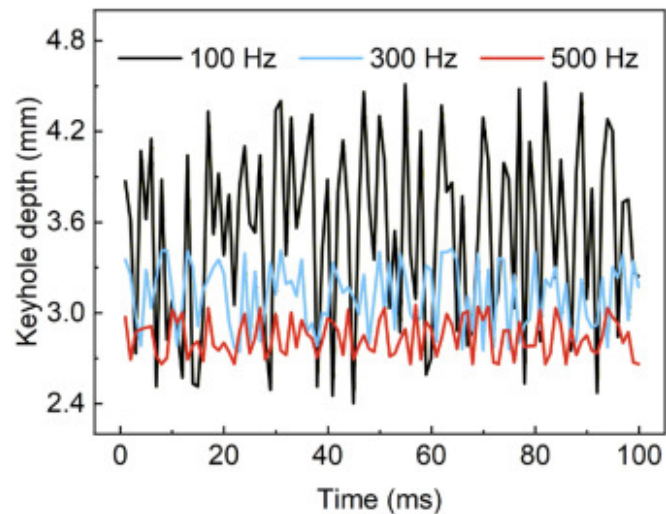
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Fig. 16. Keyhole stability without closure under high-frequency oscillating ($f = 500$ Hz, $r = 0.5$ mm).

Furthermore, since the keyhole depth changes when it becomes unstable and closes, the evolution of keyhole depth can be used to characterize its stability. Fig. 17. presents the temporal variation of keyhole depth under different oscillating frequencies. It can be observed that, under laser beam oscillation, the keyhole depth exhibits periodic fluctuations, with a period approximately

twice that of the laser beam oscillation. When the oscillating frequency increases from 100 Hz to 500 Hz, the maximum keyhole depth decreases from 4.52 mm to 3.05 mm, and the maximum depth variation reduces from 2.12 mm to 0.39 mm, corresponding to reductions of 32.5 % and 81.6 %, respectively. These results indicate that high-frequency laser beam oscillation can significantly enhance keyhole stability, mitigating the adverse effects of vortex-driven flow and droplet impact, thereby preventing keyhole collapse.



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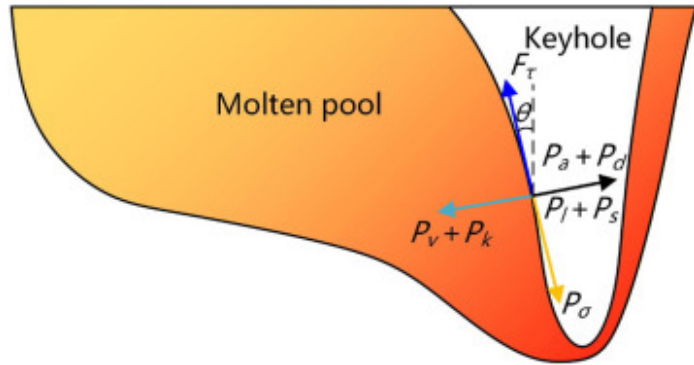
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Fig. 17. Variation of keyhole depth with laser beam oscillating frequency.

It is noteworthy that, unlike previous studies [20,25,26] which primarily relied on qualitative observations to assess keyhole stability, the present work quantitatively analyzes the depth fluctuations of the keyhole under high-frequency oscillating conditions for the first time. This provides a clearer demonstration of the stabilizing effect of beam oscillation on the keyhole and offers a solid basis for elucidating the mechanism by which beam oscillation suppresses porosity through enhanced keyhole stability. Accordingly, the enhancement of keyhole stability by beam oscillation can be considered the third type of pore-suppression mechanism.

To elucidate the mechanism underlying keyhole stability, the force characteristics of the keyhole were analyzed based on the momentum balance at its boundary. As shown in Fig. 18, keyhole stability is maintained through the interplay of multiple forces,

including metal vapor pressure P_v , arc pressure P_a , droplet impact pressure P_d , radial and tangential forces induced by surface tension (P_s and P_σ), hydrostatic pressure P_l , wall tangential shear force F_τ , and centrifugal pressure P_k generated by laser oscillating. Among these forces, P_v and P_k contribute to sustaining the keyhole opening, whereas P_a , P_d , P_s , and P_l tend to drive keyhole closure.



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Fig. 18. Force analysis model of keyhole wall in oscillating laser-arc hybrid welding.

The magnitude of the metal vapor pressure P_v can be equivalently expressed as the component of the recoil pressure acting perpendicular to the keyhole wall:

$$P_v = 0.54P_0 \exp\left(H_v \frac{T_k - T_g}{RT_k T_g}\right) \sin\theta \quad (21)$$

where P_0 is the atmospheric pressure, T_k is the temperature of the keyhole wall, T_g is the boiling point of the base metal, R is the gas constant, and θ is the angle between the keyhole wall and the vertical direction.

The droplet impact pressure P_d is generated by the momentum of the droplet impacting the molten pool, and can be expressed as:

$$P_d = \frac{4}{3} \rho_l g_d R_d \sin(\beta - \theta) \quad (22)$$

Where ρ_l denotes the density of the liquid metal, g_d is the acceleration of the droplet impacting the molten pool (taken as 32 m/s² in the experiments), R_d represents the droplet radius, and β is the arc inclination angle (55° in the experiments).

The pressure exerted on the molten pool by the periodic expansion of the arc can be expressed as [39]:

$$P_a = \frac{\mu_m I^2}{8\pi\sigma_j^2} \sin(\beta - \theta) \quad (23)$$

Where μ_m is the vacuum magnetic permeability, and σ_j is the arc heat flux density distribution coefficient (see Eq. (5) for details).

The hydrostatic pressure P_l of the liquid metal, generated by gravity, can be expressed as:

$$P_l = \rho_l g h_k \quad (24)$$

Where g is the gravitational acceleration, and h_k is the depth of the keyhole.

The surface tension P_s is given by the Young-Laplace equation:

$$P_s = \frac{2\gamma}{r_k} \quad (25)$$

Here, γ is the surface tension coefficient, and r_k is the radius of curvature of the keyhole.

The centrifugal pressure P_k arises from the stirring effect induced by the laser beam oscillation, and its expression is given by:

$$P_k = \frac{1}{2} \rho_l f^2 r^2 \quad (26)$$

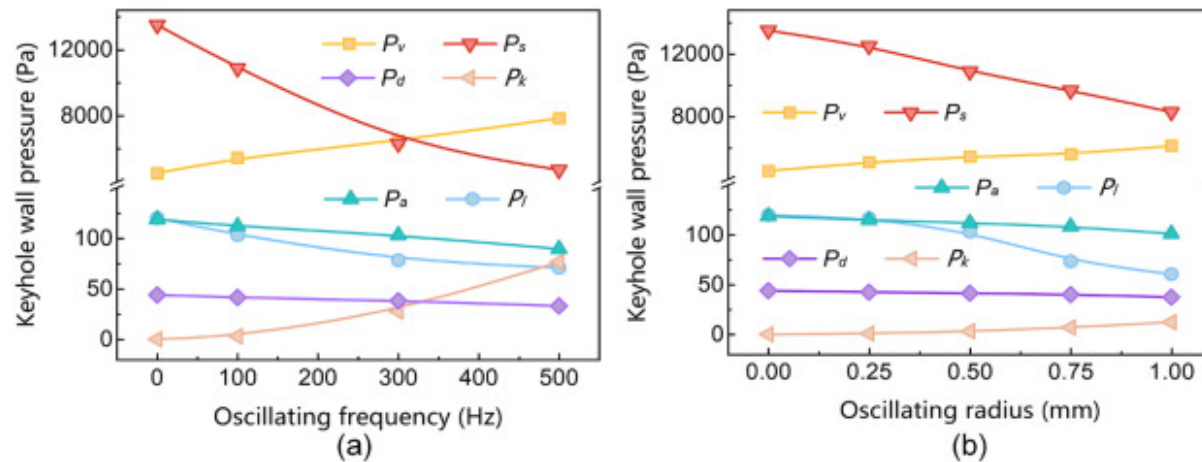
Where f and r denote the oscillating frequency and oscillating radius of the laser beam, respectively.

In summary, the resultant radial pressure P_c acting on the keyhole wall can be expressed as:

$$P_c = P_v + P_k - P_a - P_s - P_d - P_l \quad (27)$$

In the root region of the keyhole, where both the surface tension P_s and hydrostatic pressure P_l attain their maxima, the tendency for keyhole closure is greatest. Fig. 19. illustrates the variations of the forces acting on the keyhole root wall as functions

of the laser oscillating frequency f and oscillating radius r . As shown, the metal vapor pressure P_v and surface tension P_s are both on the order of kPa, significantly higher than the other radial forces, indicating that the keyhole stability is primarily governed by the dynamic balance between these two forces. When the oscillating frequency and radius are relatively low, P_v and P_k on the wall are insufficient to counteract the combined effects of P_l , P_s , P_a , and P_d , rendering the keyhole unstable and prone to closure during welding, which leads to bubble formation. As the oscillating frequency and radius increase, the stirring effect of the laser beam is enhanced, causing a simultaneous rise in P_v and P_k on the keyhole wall, while P_l , P_s , P_a , and P_d are correspondingly reduced. Consequently, the overall stability of the keyhole wall is significantly improved, effectively suppressing its closure tendency.



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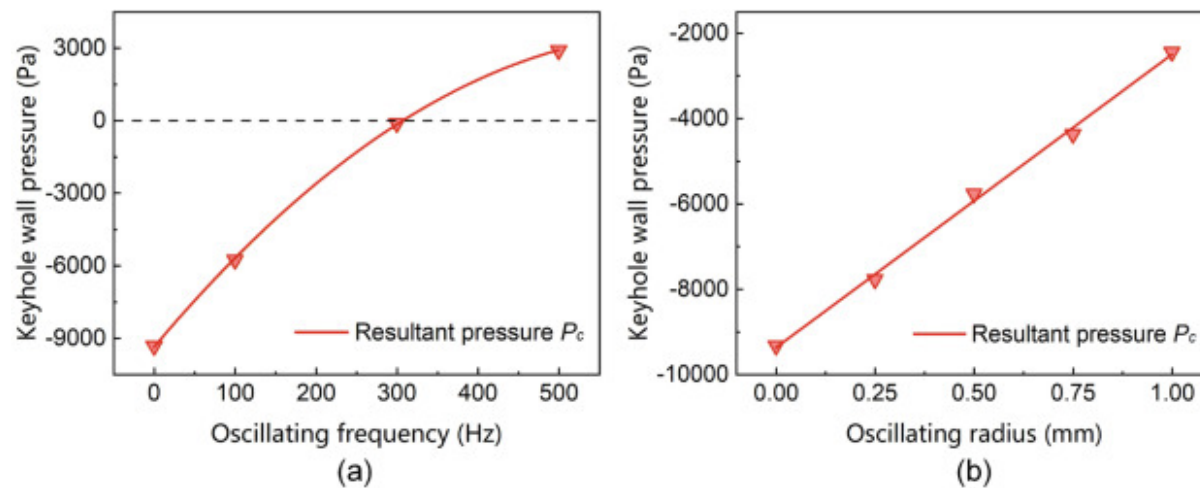
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Fig. 19. Variation of keyhole wall pressure: (a) $f = 0 \sim 500$ Hz, $r = 0.5$ mm; (b) $f = 100$ Hz, $r = 0 \sim 1.0$ mm.

Fig. 20. presents the fitted data of the resultant pressure P_c acting on the keyhole wall under different oscillating parameters, which follows the relationship given by Eq. (28). According to the momentum balance condition at the keyhole boundary, the keyhole can remain open when the radial resultant force $P_c \geq 0$. Consequently, the critical oscillating frequency at which the keyhole no longer closes is $f = 305$ Hz for a oscillating radius of $r = 0.5$ mm, while the corresponding critical oscillating radius is $r = 1.35$ mm for a oscillating frequency of $f = 100$ Hz. These results are in good agreement with the experimental findings reported by Wang et al. [23]. Moreover, compared with the work of Ke et al. [26], which only reported that beam oscillation can enhance

keyhole stability and reduce the tendency for keyhole closure, the present study goes a step further by establishing a quantitative mathematical model based on a force analysis of the keyhole wall. This model enables direct evaluation of the influence of oscillating parameters on keyhole stability and allows accurate prediction of the critical oscillating conditions required to prevent keyhole closure. The proposed model not only provides scientific guidance for process optimization and weld quality improvement in oscillating laser-arc hybrid welding, but also represents a significant advancement over existing studies in terms of theoretical quantification and practical applicability.

$$\begin{cases} P_{cf} = -0.031f^2 + 39.96f - 9301.92 \\ P_{cr} = 6898.07r - 9305.78 \end{cases} \quad (28)$$



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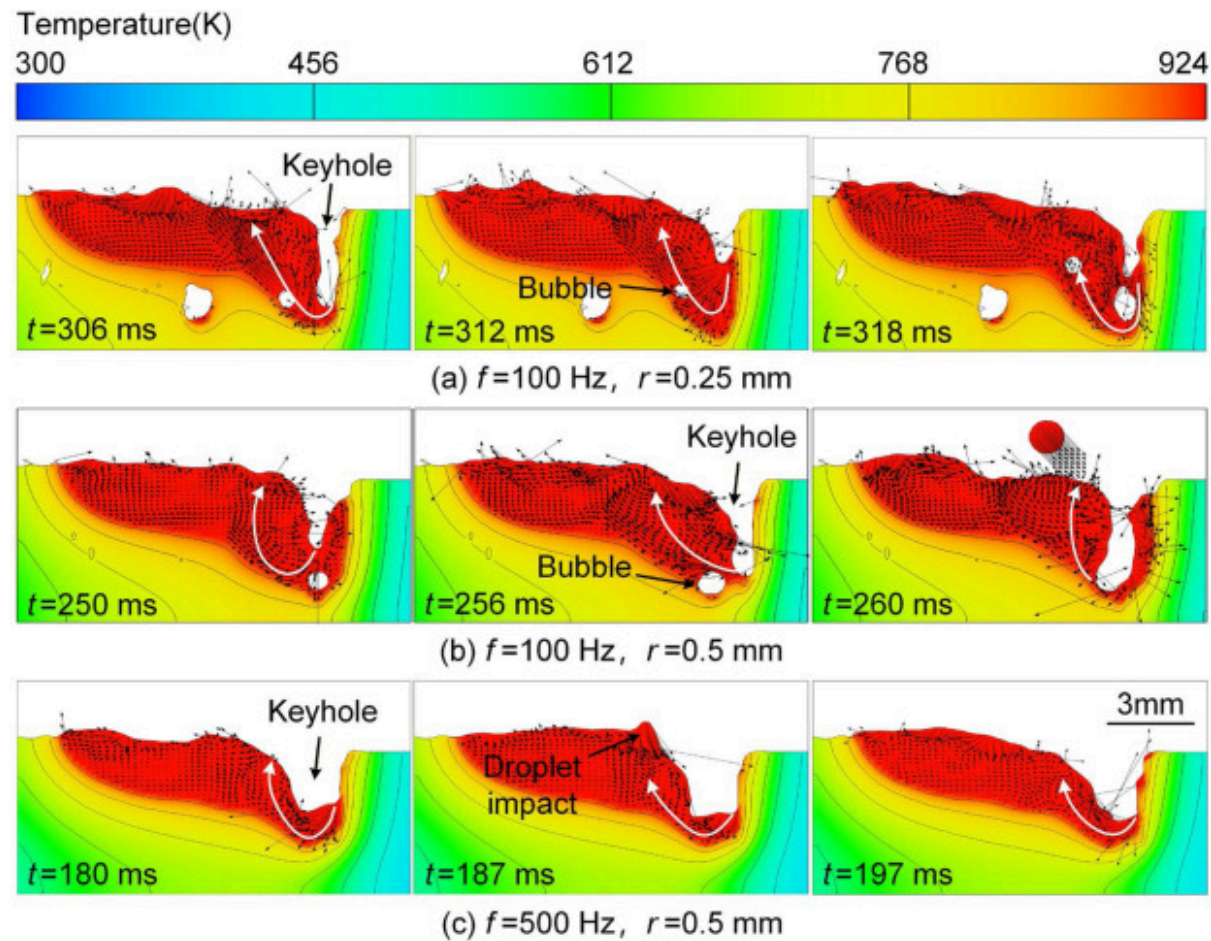
Fig. 20. Keyhole wall contact pressure P_c : (a) $f = 0 \sim 500$ Hz, $r = 0.5$ mm; (b) $f = 100$ Hz, $r = 0 \sim 1.0$ mm.

3.5. Mechanisms of porosity suppression in oscillating laser arc hybrid welding

As revealed by the preceding analysis, the suppression mechanism of laser beam oscillation against porosity primarily involves three pathways: (1) accelerating bubble escape from the molten pool, (2) absorbing bubbles through dynamic keyholes, and (3)

enhancing keyhole stability via beam oscillation. However, as oscillating frequency and radius increase, the suppression mechanisms also exhibit corresponding changes.

When the oscillating frequency and radius are relatively low (e.g., $f = 100$ Hz, $r = 0.25$ mm), the laser beam primarily suppresses porosity by enhancing fluidity within the molten pool and accelerating bubble migration and ascent (Fig. 21a). In this process, the clockwise vortex formed behind the keyhole acts as a momentum source, driving bubbles to ascend rapidly and thereby reducing the likelihood of its capture by the solid-liquid interface and subsequent pore formation. When the oscillating radius r increases to 0.5 mm (Fig. 21b), the laser-induced rotation of the keyhole cover a larger area, enabling interactions between the keyhole and bubbles and facilitating their coalescence and absorption, which indirectly suppresses pore formation. Furthermore, when the oscillating frequency is increased to $f = 500$ Hz (Fig. 21c), high-frequency oscillating significantly enhances the morphological stability of the keyhole, allowing it to effectively resist disturbances induced by droplet impacts, remain continuously open, and prevent keyhole collapse and bubble formation, thereby achieving complete suppression of porosity at its source.



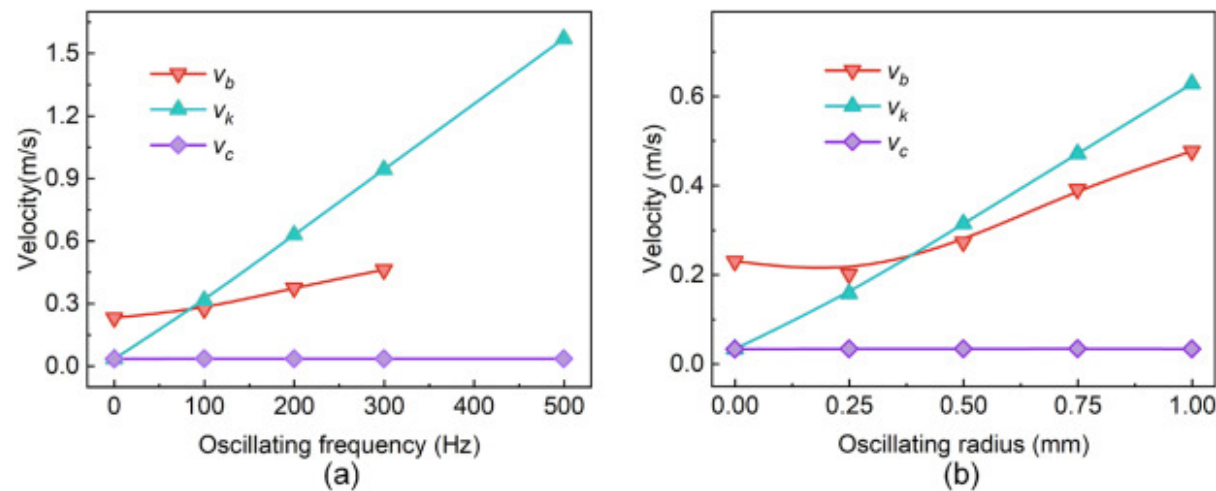
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Fig. 21. Porosity suppression by laser beam oscillation: (a) accelerating bubble escape from the molten pool; (b) dynamic keyhole-induced bubble coalescence and absorption; (c) enhancement of keyhole stability to prevent closure.

To further elucidate the competitive interplay among the three porosity-suppression mechanisms, this study investigates the variations of bubble migration velocity v_b , laser beam oscillating velocity v_k , and molten-pool solidification velocity v_c under different oscillating parameters based on the thermofluidic analysis of the molten pool (Fig. 22). It can be seen that when $f = 100$ Hz and $r < 0.5$ mm, the bubble migration velocity v_b exceeds both the laser oscillating velocity v_k and the molten pool

solidification velocity v_c . In this scenario, the likelihood of interaction with the laser keyhole is relatively low. Instead, the laser beam primarily enhances molten pool flow via the oscillatory stirring effect, accelerating bubble rise and thereby reducing the probability of pore formation in the weld (see Fig. 13). As the oscillating frequency and radius increase, both v_b and v_k increase, while v_c remains essentially constant. When $f = 100$ Hz and $r > 0.5$ mm, the laser oscillating velocity v_k exceeds the bubble migration velocity v_b ; in this case, the laser primarily suppresses porosity by guiding the keyhole in a rotational motion, enabling it to encounter and coalesce with bubbles, thereby reducing pore formation (see Fig. 15). Furthermore, as shown in Fig. 20, when $f > 305$ Hz, $r = 0.5$ mm or $f = 100$ Hz, $r \geq 1.35$ mm, the net pressure on the keyhole wall $P_c \geq 0$, ensuring that the keyhole remains continuously open without collapse, thereby fundamentally preventing porosity formation (as illustrated in Fig. 16.). Under this condition, the enhancement of keyhole stability induced by laser beam oscillation becomes the dominant mechanism for porosity suppression.



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Fig. 22. Bubble migration velocity v_b , laser beam oscillating velocity v_k , and molten pool solidification velocity v_c : (a) $f = 0 \sim 500$ Hz, $r = 0.5$ mm; (b) $f = 100$ Hz, $r = 0 \sim 1.0$ mm.

Based on the above conclusions, the previous experimental results (see Fig. 6.) can be reasonably interpreted. Under low-frequency and small-radius oscillating conditions (e.g., $f = 100$ Hz, $r < 0.5$ mm), porosity suppression primarily relies on the molten pool stirring effect induced by laser beam oscillation, which accelerates bubble migration and escape. However, this

mechanism has a limited effect, and the weld still exhibits a relatively high porosity. As the oscillating frequency and radius increase, the dominant porosity suppression mechanism gradually shifts to the dynamic keyhole-mediated coalescence and absorption of bubbles. Compared with the former, this mechanism more effectively reduces bubble retention, thereby significantly lowering the weld porosity. When oscillating parameters meet the critical condition for $P_c \geq 0$ on the keyhole wall (e.g., $f = 500$ Hz, $r = 0.5$ mm), the keyhole remains highly stable throughout the welding process without collapse. Under these conditions, pore formation is fundamentally suppressed, resulting in a weld with zero porosity.

4. Conclusion

- (1) With increasing oscillating frequency f and oscillating radius r , the weld porosity is significantly reduced. As r increases from 0 to 1.0 mm, the porosity decreases from 6.64 % to 0.61 %, corresponding to a 90.8 % reduction. When f increases from 0 to 500 Hz, the porosity decreases from 6.64 % to 0, achieving complete elimination of pores. Further analysis indicates that full suppression of weld porosity can be realized when the oscillating parameters satisfy $f \geq 300$ Hz, $r \geq 0.75$ mm or $f \geq 500$ Hz, $r \geq 0.5$ mm.
- (2) Vortex flow and droplet impact are the primary factors leading to keyhole instability, closure, and bubble formation. Specifically, vortex flow promotes bubble generation by closing the bottom of the keyhole, whereas droplet impact induces bubble formation by compressing and closing the keyhole. If the generated bubbles fail to escape in a timely manner during subsequent migration, they are trapped by the solid-liquid interface of the molten pool and eventually evolve into pores.
- (3) The porosity suppression mechanisms in OLAHW can be primarily categorized into three types: (i) the laser beam oscillation drives vortex flow, promoting bubble rise and escape from the molten pool; (ii) the laser beam guides the keyhole in a rotational motion to facilitate bubble coalescence and absorption; and (iii) laser beam oscillation enhances keyhole stability, preventing keyhole closure and subsequent bubble formation.
- (4) The porosity suppression mechanism of laser beam oscillation in OLAHW depends on the competitive interplay among the three porosity suppression modes under different oscillating parameters. Specifically, When $f = 100$ Hz and $r < 0.5$ mm, vortex-driven bubble evacuation predominates, accelerating bubble ascent and escape to reduce pore formation. When $f = 100$ Hz and $r \geq 0.5$ mm, keyhole rotation becomes the

dominant mechanism, effectively suppressing porosity by capturing and coalescing bubbles. Finally, When $f \geq 305$ Hz, $r = 0.5$ mm or $f = 100$ Hz, $r \geq 1.35$ mm, the enhancement of keyhole stability induced by beam oscillation becomes the dominant suppression mechanism. Under these conditions, the compressive pressure on the keyhole wall remains positive, preventing keyhole collapse and bubble generation, thereby eliminating porosity at its source.

CRediT authorship contribution statement

Jianeng Xu: Conceptualization, Methodology, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Suning Zhao: Conceptualization, Supervision, Funding acquisition, Project administration, Investigation. **Laihege Jiang:** Methodology, Data curation. **Ling Cen:** Methodology, Software. **Lin Shi:** Methodology, Data curation. **Ming Gao:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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